



Lecture No. 2





Introduction to Fuzzy Logic



What is the difference between Fuzzy Logic and Neural Networks?

- Fuzzy logic allows making definite decisions based on imprecise or ambiguous data.
- ANN tries to incorporate human thinking process to solve problems without mathematically modeling them.
- Both these methods can be used to solve nonlinear problems, and problems that are not properly specified, but they are not related.
- ANN tries to apply the thinking process in the human brain to solve problems.

SC Constituents and Conventional AI

- “SC is an emerging approach to computing which parallel the remarkable ability of the human mind to reason and learn in a environment of uncertainty and imprecision” [Lotfi A. Zadeh, 1992]
- SC consists of several computing paradigms including:
 - NN
 - Fuzzy set theory
 - Approximate reasoning
 - Derivative-free optimization methods such as genetic algorithms (GA) & simulated annealing (SA)

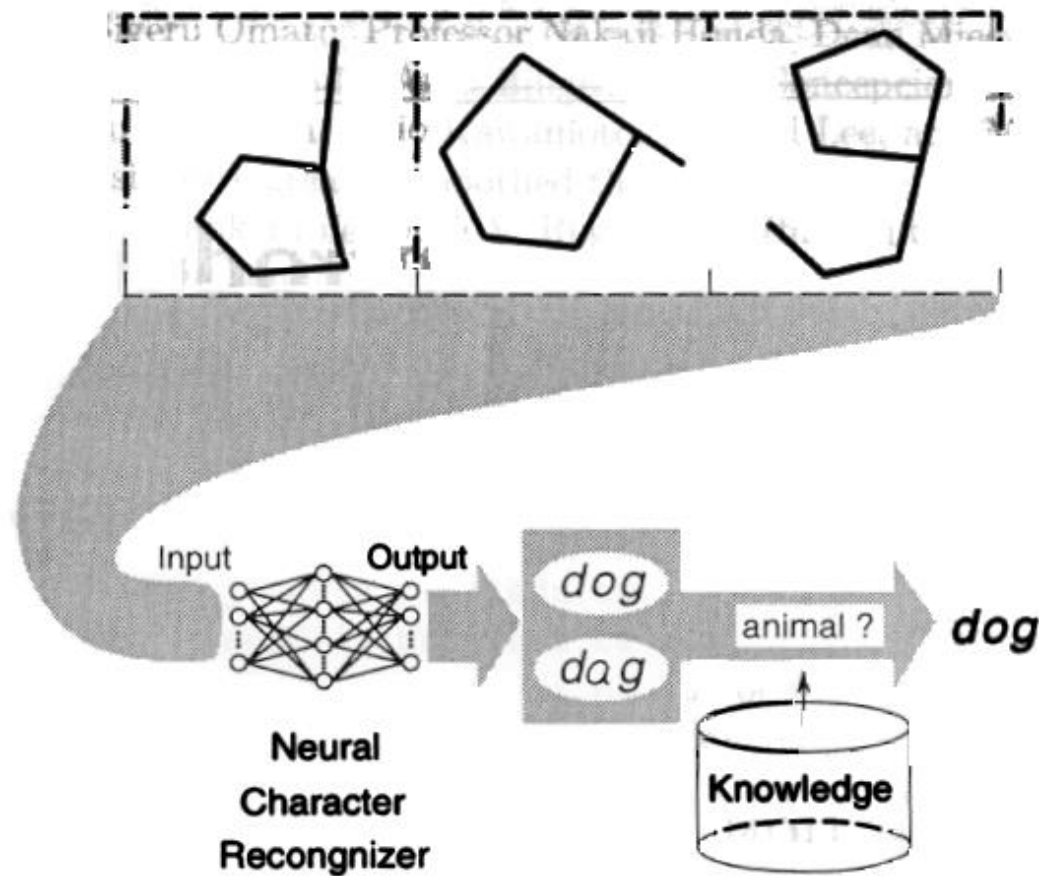
SC constituents (the first three items) and conventional AI

Methodology	Strength
Neural network	Learning and adaptation
Fuzzy set theory	Knowledge representation via fuzzy if-then rules
Genetic algorithm and simulated annealing	Systematic random search
Conventional AI	Symbolic manipulation

Contd....

- These methodologies form the core of SC
- In general, SC does not perform much symbolic manipulation
- SC in this sense complements conventional AI approaches

Character recognizer



Features of Conventional AI

- From conventional AI to computational intelligence
 - Conventional AI manipulates symbols on the assumption that human intelligence behavior can be stored in symbolically structured knowledge bases: this is known as: “ The physical symbol system hypothesis
 - The knowledge-based system (or expert system) is an example of the most successful conventional AI product

NF and SC characteristics

- With NF modeling as a backbone, SC can be characterized as:
 - Human expertise (fuzzy if-then rules)
 - Biologically inspired computing models (NN)
 - New optimization techniques (GA, SA, RA)
 - Numerical computation (no symbolic AI so far, only numerical)

NF and SC Characteristics Contd...

- New application domains: mostly computation intensive like adaptive signal processing, adaptive control, nonlinear system identification etc
- Model free learning:-models are constructed based on the target system only
- Intensive computation: based more on computation

NF and SC Characteristics Contd...

- Fault tolerance: deletion of a neuron or a rule does not destroy the system. The system performs with lesser quality
- Goal driven characteristics:- only the goal is important and not the path.
- Real world application:- large scale, uncertainties

WHAT IS FUZZY LOGIC?

○ Definition of fuzzy

- Fuzzy – “not clear, distinct, or precise; blurred”

○ Definition of fuzzy logic

- A form of knowledge representation suitable for notions that cannot be defined precisely, but which depend upon their contexts.

TRADITIONAL REPRESENTATION OF LOGIC



Slow

Speed = 0



Fast

Speed = 1

```
bool speed;  
get the speed  
if ( speed == 0) {  
    // speed is slow  
}  
else {  
    // speed is fast  
}
```

FUZZY LOGIC REPRESENTATION

- For every problem must represent in terms of fuzzy sets.



Slowest

[0.0 – 0.25]

- What are fuzzy sets?



Slow

[0.25 – 0.50]



Fast

[0.50 – 0.75]



Fastest

[0.75 – 1.00]

FUZZY LOGIC REPRESENTATION CONT.



Slowest

Slow

Fast

Fastest

```
float speed;  
get the speed  
if ((speed >= 0.0)&&(speed < 0.25)) {  
    // speed is slowest  
}  
else if ((speed >= 0.25)&&(speed < 0.5))  
{  
    // speed is slow  
}  
else if ((speed >= 0.5)&&(speed < 0.75))  
{  
    // speed is fast  
}  
else // speed >= 0.75 && speed < 1.0  
{  
    // speed is fastest  
}
```

ORIGINS OF FUZZY LOGIC

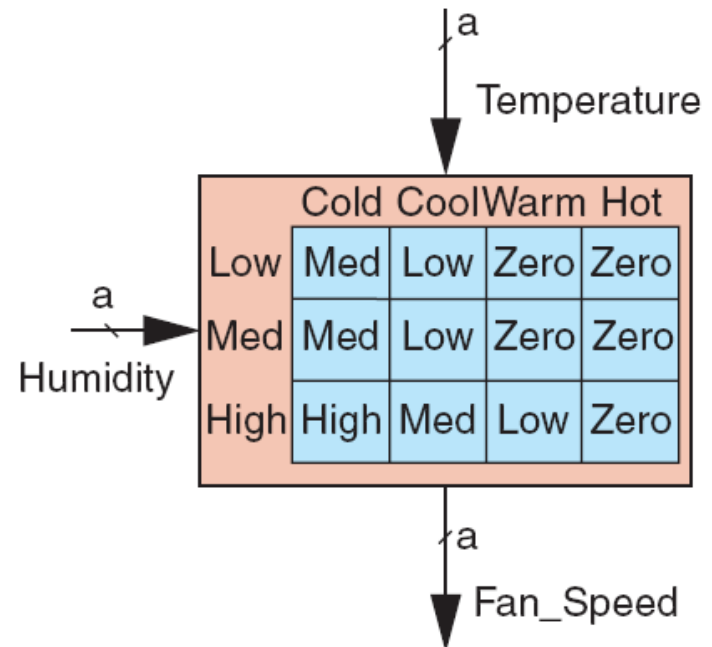
- Traces back to Ancient Greece
- Lotfi Asker Zadeh (1965)
 - First to publish ideas of fuzzy logic.
- Professor Toshire Terano (1972)
 - Organized the world's first working group on fuzzy systems.
- F.L. Smidth & Co. (1980)
 - First to market fuzzy expert systems.

FUZZY LOGIC IN CONTROL SYSTEMS

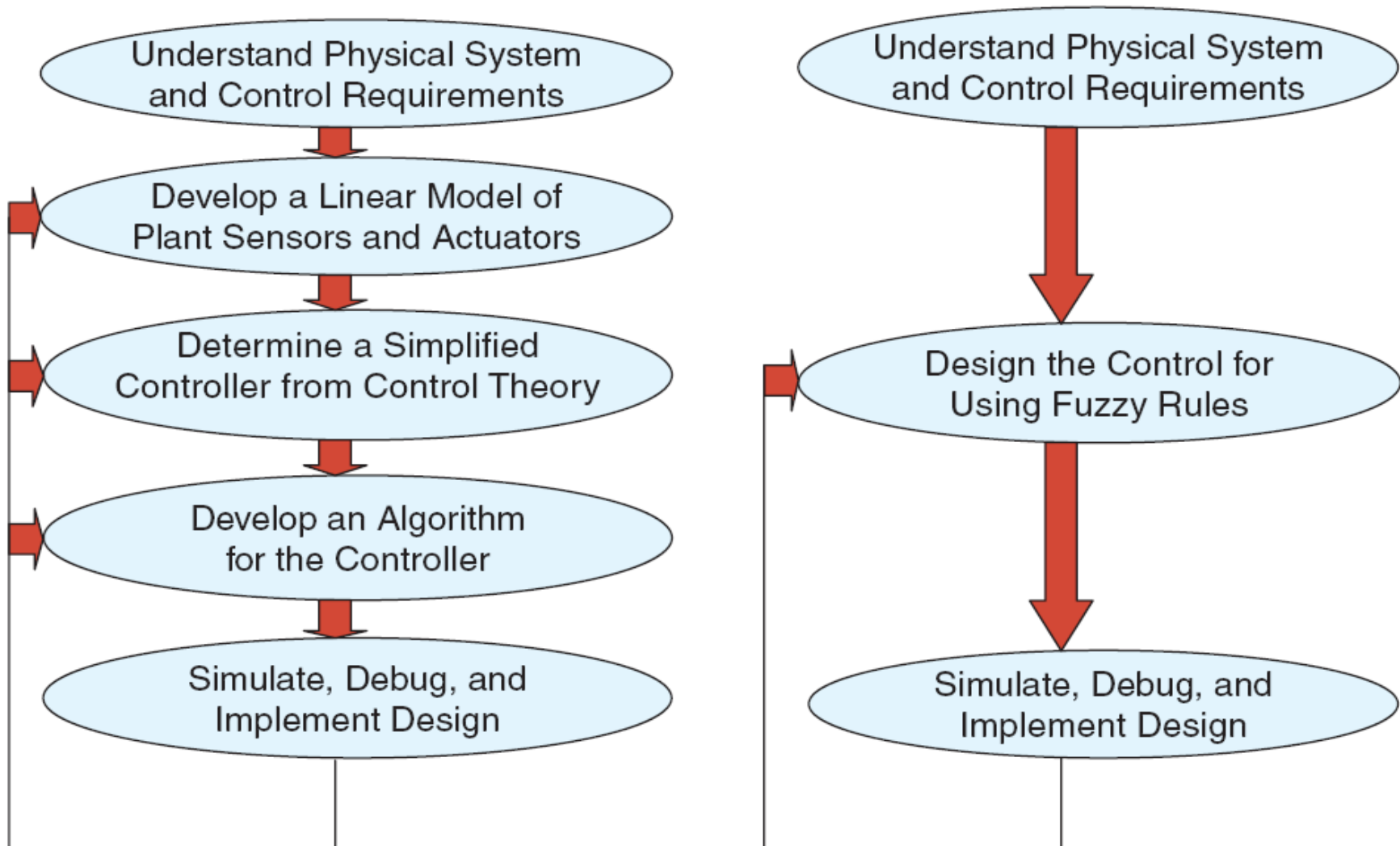
- Fuzzy Logic provides a more efficient and resourceful way to solve Control Systems.
- Some Examples
 - Temperature Controller
 - Anti – Lock Break System (ABS)

TEMPERATURE CONTROLLER

- The problem
 - Change the speed of a heater fan, based off the room temperature and humidity.
- A temperature control system has four settings
 - Cold, Cool, Warm, and Hot
- Humidity can be defined by:
 - Low, Medium, and High
- Using this we can define the fuzzy set.



BENEFITS OF USING FUZZY LOGIC



ANTI LOCK BREAK SYSTEM (ABS)

- Nonlinear and dynamic in nature
- Inputs for Intel Fuzzy ABS are derived from
 - Brake
 - 4 WD
 - Feedback
 - Wheel speed
 - Ignition
- Outputs
 - Pulsewidth
 - Error lamp

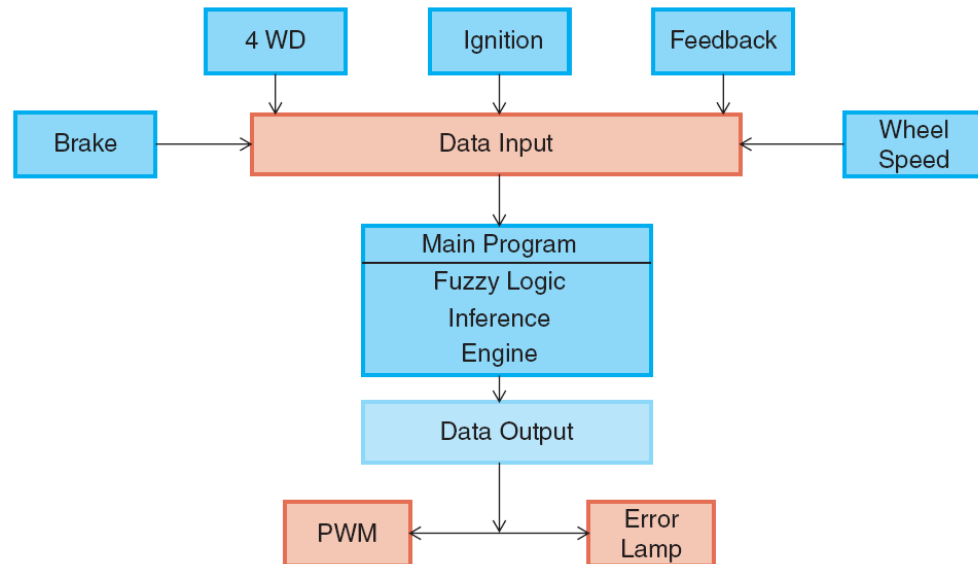


Fig. 6 ABS block diagram

FUZZY LOGIC IN OTHER FIELDS

- Business
- Hybrid Modeling
- Expert Systems

Summary

- Fuzzy logic provides an alternative way to represent linguistic and subjective attributes of the real world in computing.
- It is able to be applied to control systems and other applications in order to improve the efficiency and simplicity of the design process.
- SC is evolving rapidly
- New techniques and applications are constantly being proposed

Fuzzy set theory

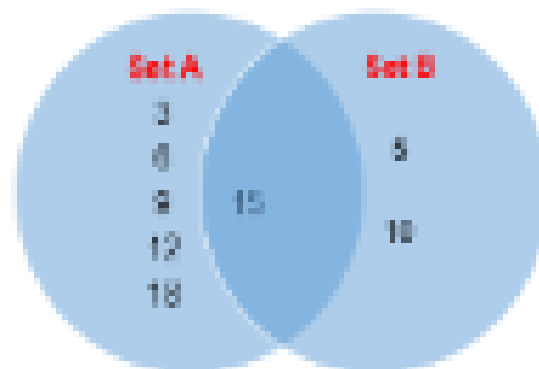
- Human brains interpret imprecise and incomplete sensory information provided by perceptive organs
- Fuzzy set theory provides a systematic calculus to deal with such information linguistically
- It performs numerical computation by using linguistic labels stimulated by membership functions
- It lacks the adaptability to deal with changing external environments ==> incorporate NN learning concepts in fuzzy inference systems: NF modeling

Evolutionary computation

- Natural intelligence is the product of millions of years of biological evolution
- Simulation of complex biological evolutionary processes
- GA is one computing technique that uses an evolution based on natural selection
- Immune modeling and artificial life are similar disciplines based on chemical and physical laws
- GA and SA population-based systematic random search (RA) techniques

Brief Review of convention Sets

- Definition:
 - Sets: A collection of objects having one or more common characteristics.
- Members/Elements: Objects belonging to a set is represented as $A \in P$, where P is a set



Brief Review of convention Sets

Subset: BP_2 is said to be a subset of set AP_1 if only if $B \in P_2 \Rightarrow B \in P_1, \forall_B$. This sounds as $B \subseteq A$.

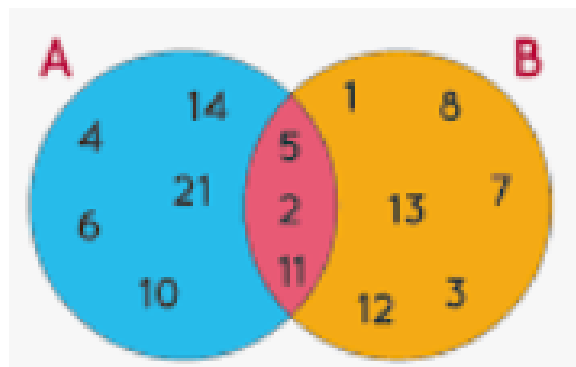
Proper Subset: A set BP_2 could be a proper subset of set AP_1 if only and only if P_2 span is a subset of P_1 and $\exists_A \in P_1$, such that $A \notin P_2$, so then, is written as $B \subset A$.

Sets number operations

Equal Sets: Two sets A and B are said to be equal iff $\forall x \in A$ and $\forall y \in B, x = y$.

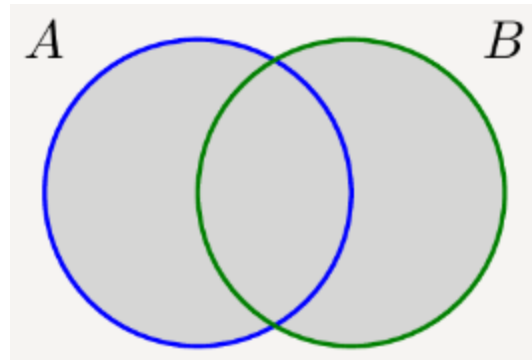


Intersection Operation: For any two sets A and B if $\exists x$ common in both A and B , then $x \in (A \cap B)$ where $\cap$ denotes the logical intersection operation.

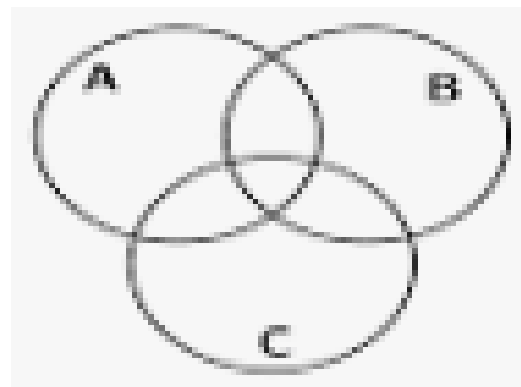


Sets number operations

Union Operation: For any two sets A and B if $\exists_x$ which is a member of either A or B then $x \in (A \cup B)$ where $\cup$ denotes the logical union operation.



Universal Set: A universal set U is a set that has all possible numbers of a particular domain.



Concept of a Fuzzy membership

Zero



Almost zero



Near zero



fuzzy sets Vs Classical set

Every member x of a fuzzy set, A , is assigned a fuzzy index $\mu_A(x)$ in the universal of $(0,1)$ which is often called as the grade of membership of x in A .

In classical set, membership grade $\mu_A(x)$ is either 0 or 1.

Fuzzy sets: A is a set of ordered pairs given by

$A = ([x, \mu_A(x)] : x \in X)$ where X is universal set

and $\mu_A(x)$ is the grade of membership of the object x in A .

usually $\mu_x(x)$ lies in $(0, 1)$.

Membership Functions: A membership function $\mu_x(x)$ is characterized by, $\mu_x: x \rightarrow [0,1], x \in X$

Where x is a real number describing on object or its attribute, X , is the universal of discourse and ACX .

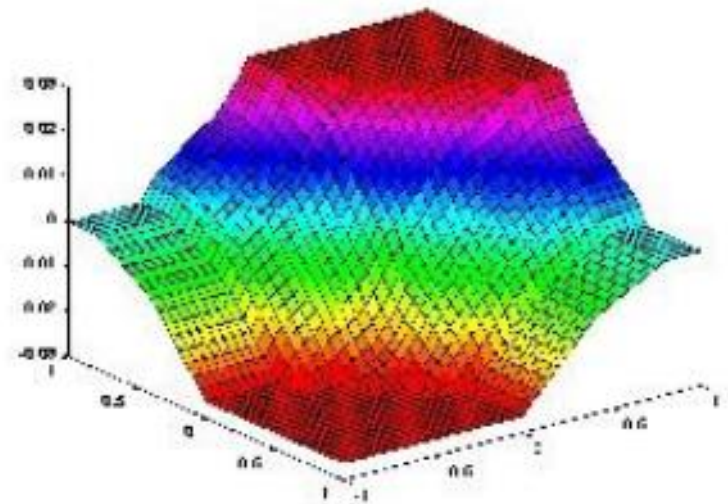
Main characteristics

Fuzzy sets:

precise model in a finite number of points, smooth transition (approximation) among them.

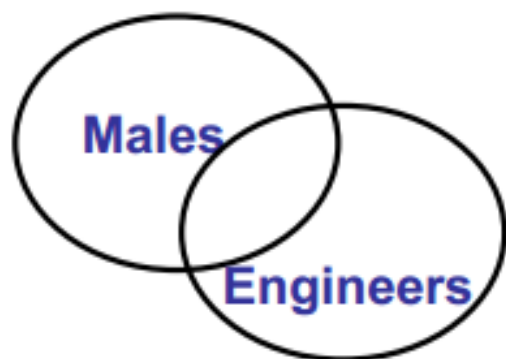
E.g.: control of a power plant

We can define what to do at the regimen (e.g., steam temperature = 120°, steam pressure 3 atm), and when in critical situations (e.g., steam temperature = 100°), and design a model that smoothly goes from one point to the other.

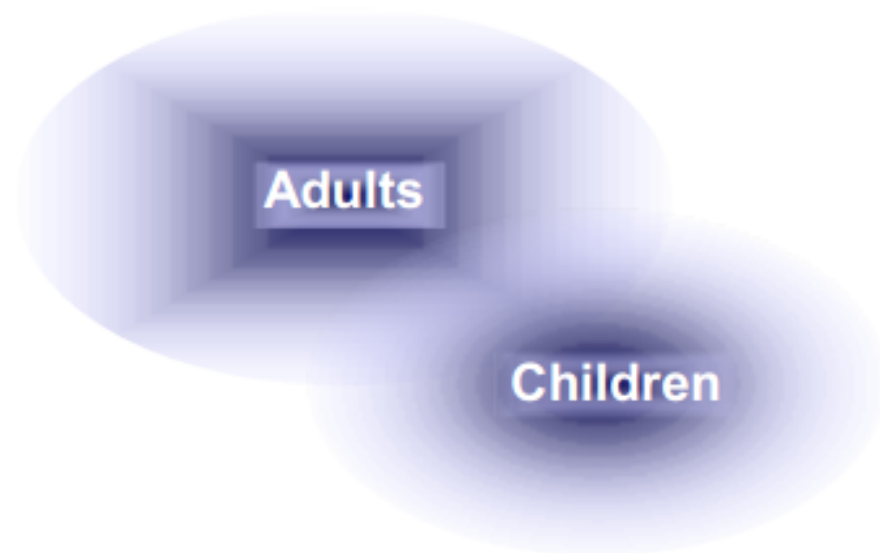


What is a fuzzy set?

A fuzzy set is a set whose membership function may range on the interval $[0,1]$.



Crisp sets

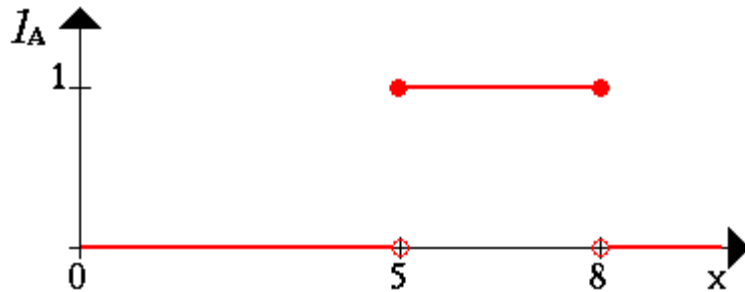


Fuzzy sets

crisp sets

- **Boolean/Crisp set A** is a mapping for the elements of S to the set $\{0, 1\}$, i.e., $A: S \rightarrow \{0, 1\}$
- *Characteristic function:*
- $\mu_A(x) = \begin{cases} 1 & \text{if } x \text{ is an element of set } A \\ 0 & \text{if } x \text{ is not an element of set } A \end{cases}$
- **Fuzzy set F** is a mapping for the elements of S to the interval $[0, 1]$, i.e., $F: S \rightarrow [0, 1]$
- Characteristic function: $0 \leq \mu_F(x) \leq 1$
- 1 means full membership, 0 means no membership and anything in between, e.g., 0.5 is called **graded membership**

Definition of a fuzzy set



Classical set A in X is a set of ordered pairs

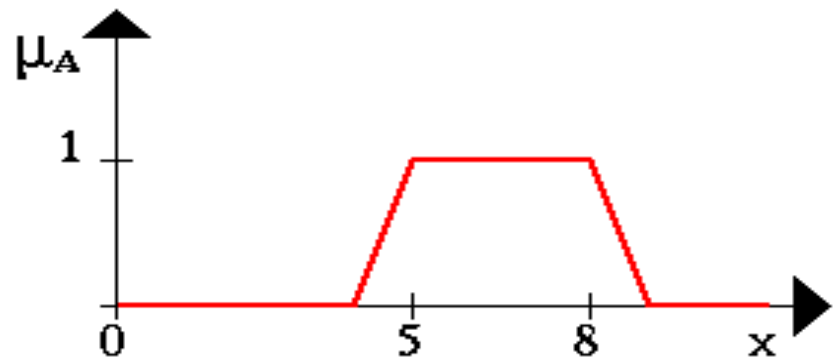
$$A = \{(x, I_A(x)) \mid x \in X\},$$

defined by **indicator function** $I_A(x) \in \{0, 1\}$

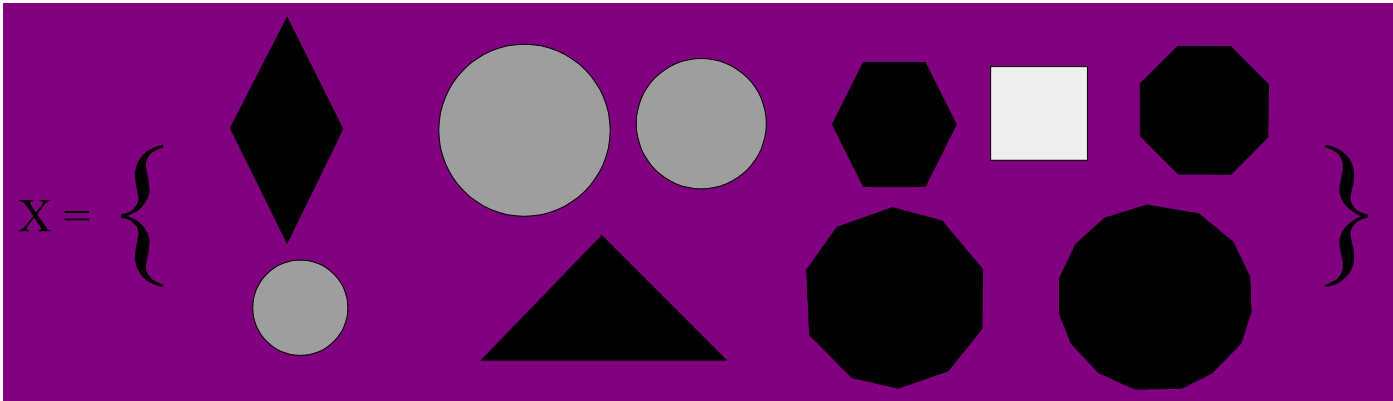
Fuzzy set A in X is as a set of ordered pairs

$$A = \{(x, \mu_A(x)) \mid x \in X\},$$

defined by **membership function** $0 \leq \mu_A(x) \leq 1$



Crisp and Fuzzy example



One can define the crisp set “circles” as:

$$C = \{ \text{small light gray circle}, \text{medium light gray circle}, \text{large light gray circle} \}$$

The fuzzy set “circles can be defined as:

$$C = \{ (\text{black hexagon}, 0.1), (\text{black octagon}, 0.3), (\text{black decagon}, 0.5), (\text{large black circle}, 0.8), \\ (\text{small light gray circle}, 1.0), (\text{medium light gray circle}, 1.0), (\text{large light gray circle}, 1.0) \}$$

Fuzzy Linguistic variables

Examples of fuzzy measures include:

close, heavy, light, big, small, smart, fast, slow, hot, cold, Tall, Short,

- Fuzzy sets and concepts are commonly used in natural language

*John is **tall***

*Dan is **smart***

*Alex is **happy***

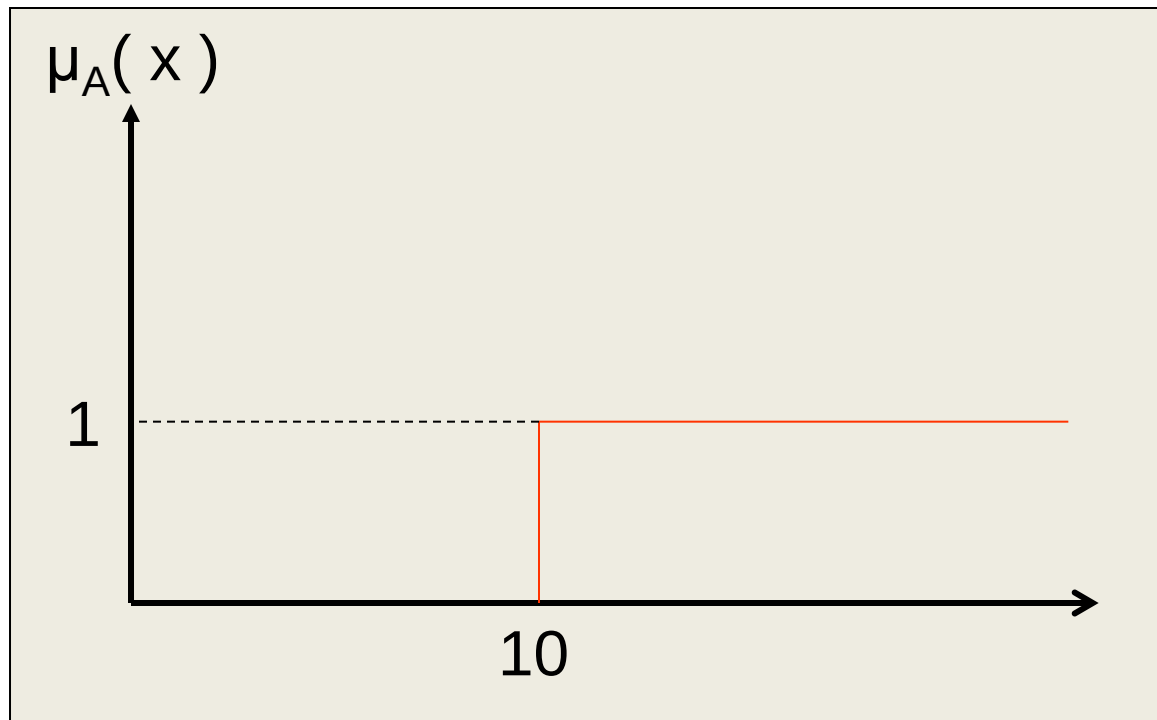
*The class is **hot***

Crisp membership function

Crisp membership functions (μ) are either one or zero.

- e.g. Numbers greater than 10.

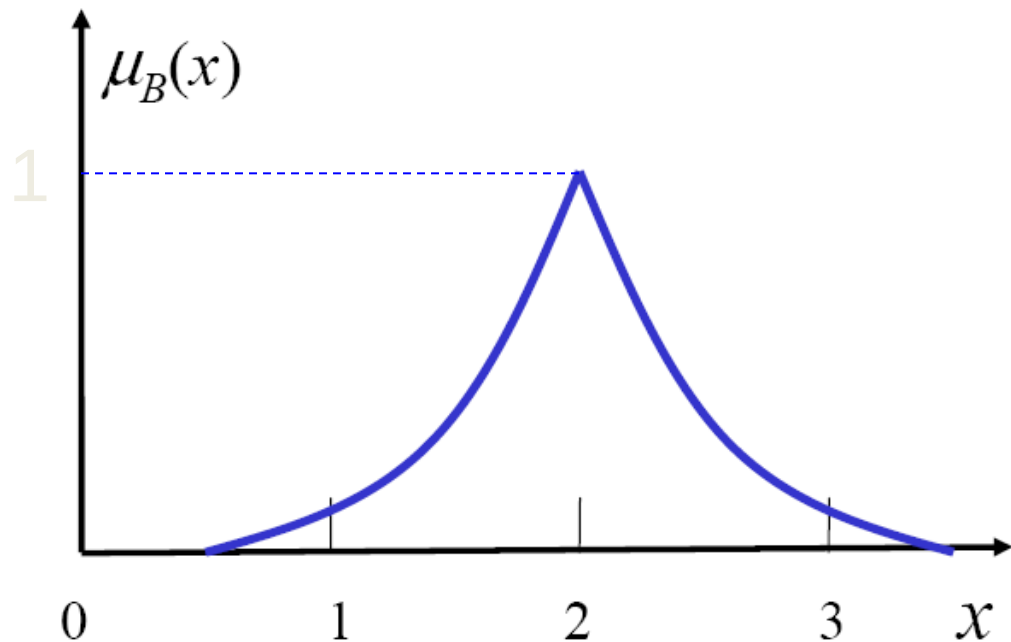
$$A = \{x \mid x > 10\}$$



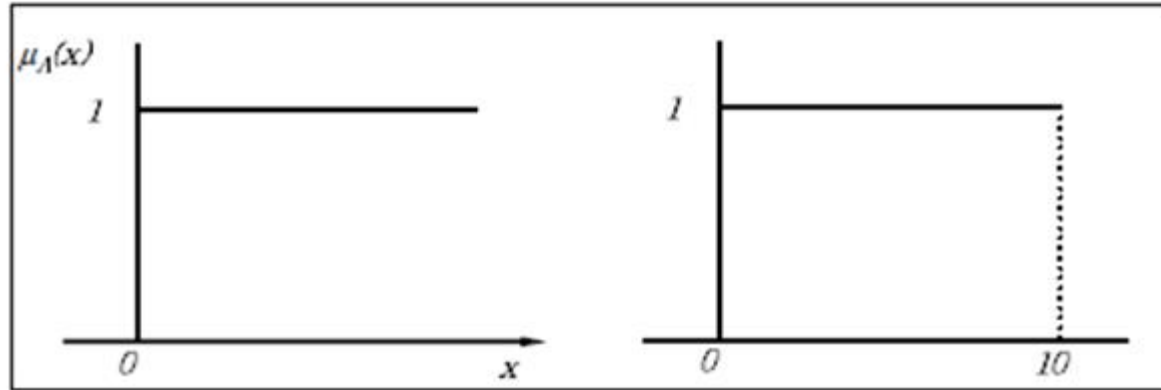
Fuzzy membership function

The set, B , of numbers *near* to 2 can be represented by a membership function:

$$\mu_B(x) = e^{-|x-2|}$$

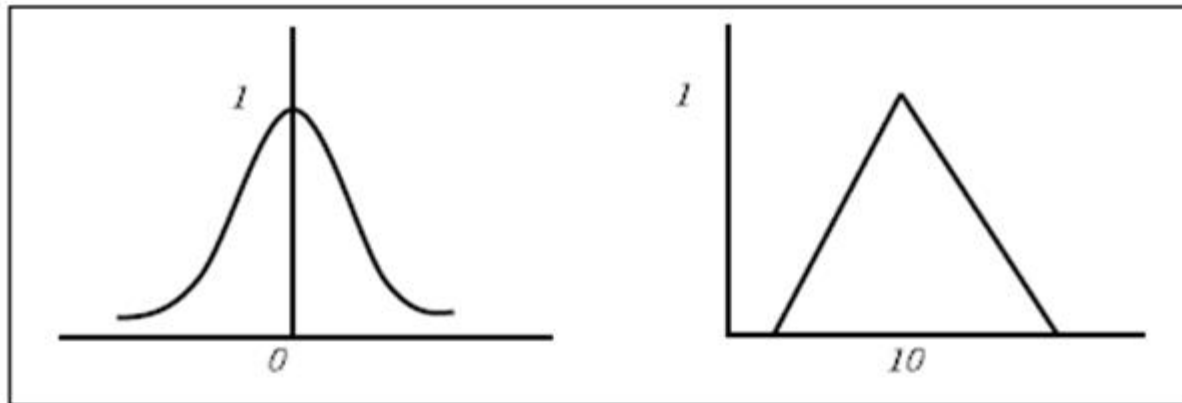


Fuzzy membership function



Set “positive number”

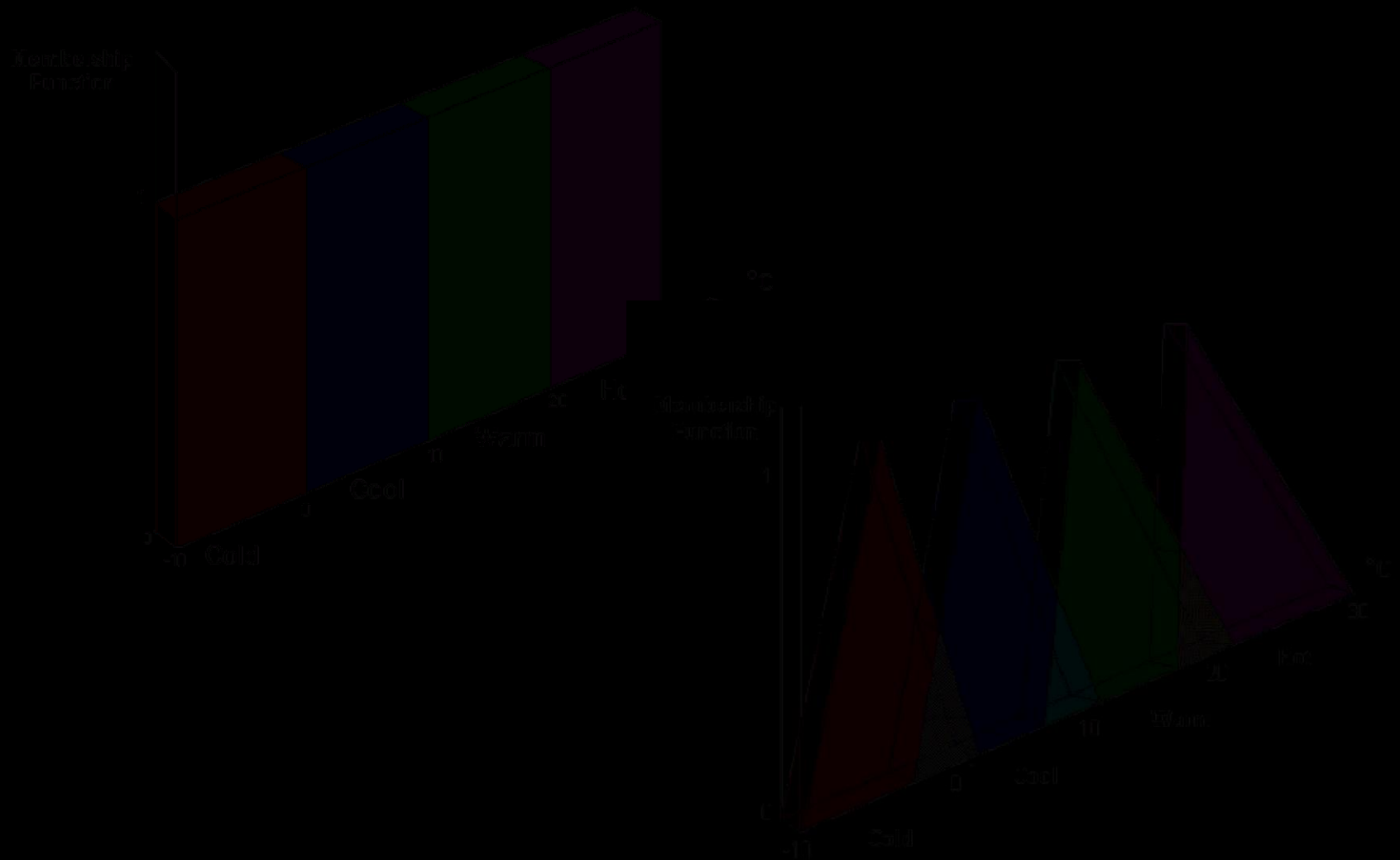
Set “positive number
not exceeding 10”



Set “number near 0”

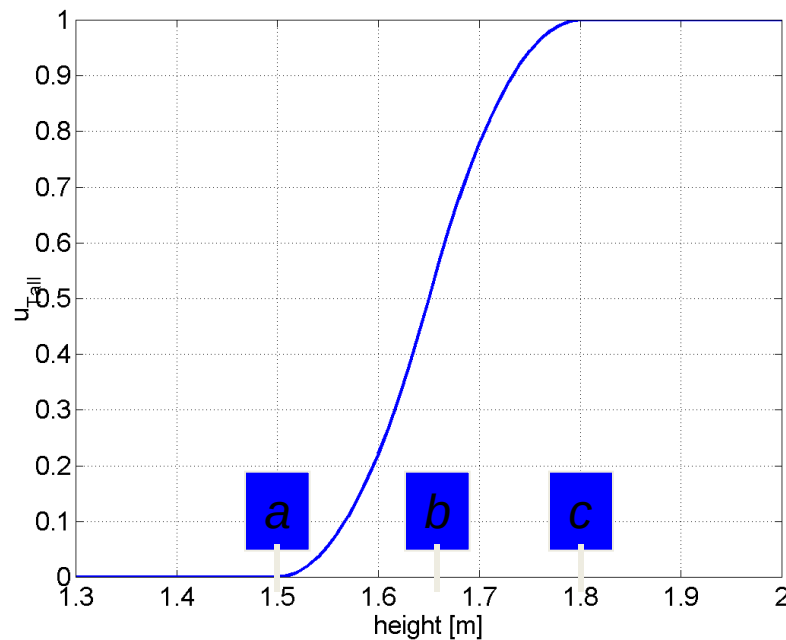
Set “number near 10”

Crisp & Fuzzy Membership functions



Fuzzy Membership functions: S-function

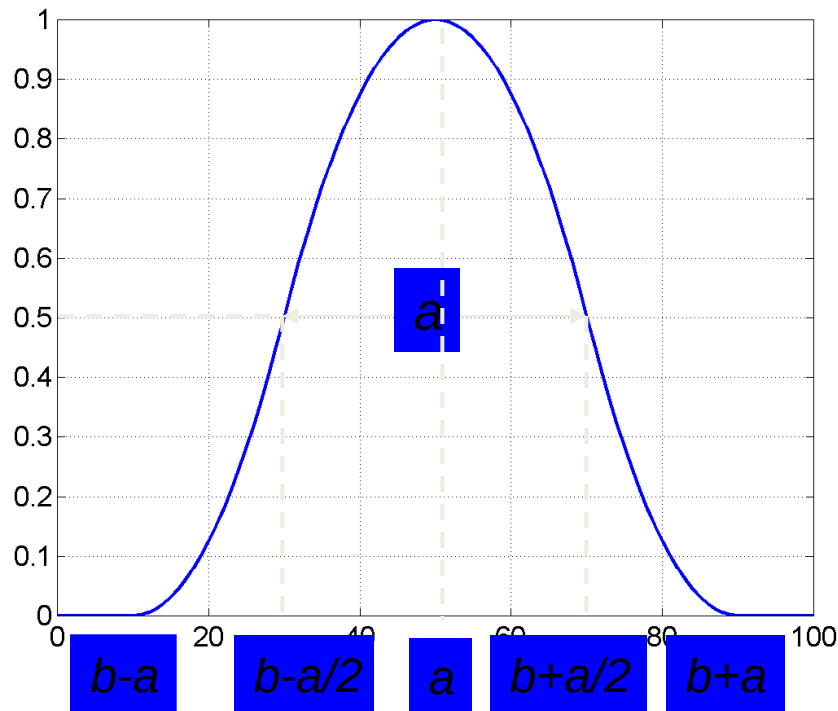
- The S-function can be used to define fuzzy sets
- $S(x, a, b, c) =$
 - 0 for $x \leq a$
 - $2(x-a/c-a)^2$ for $a \leq x \leq b$
 - $1 - 2(x-c/c-a)^2$ for $b \leq x \leq c$
 - 1 for $x \geq c$



Fuzzy Membership functions: Π -Function

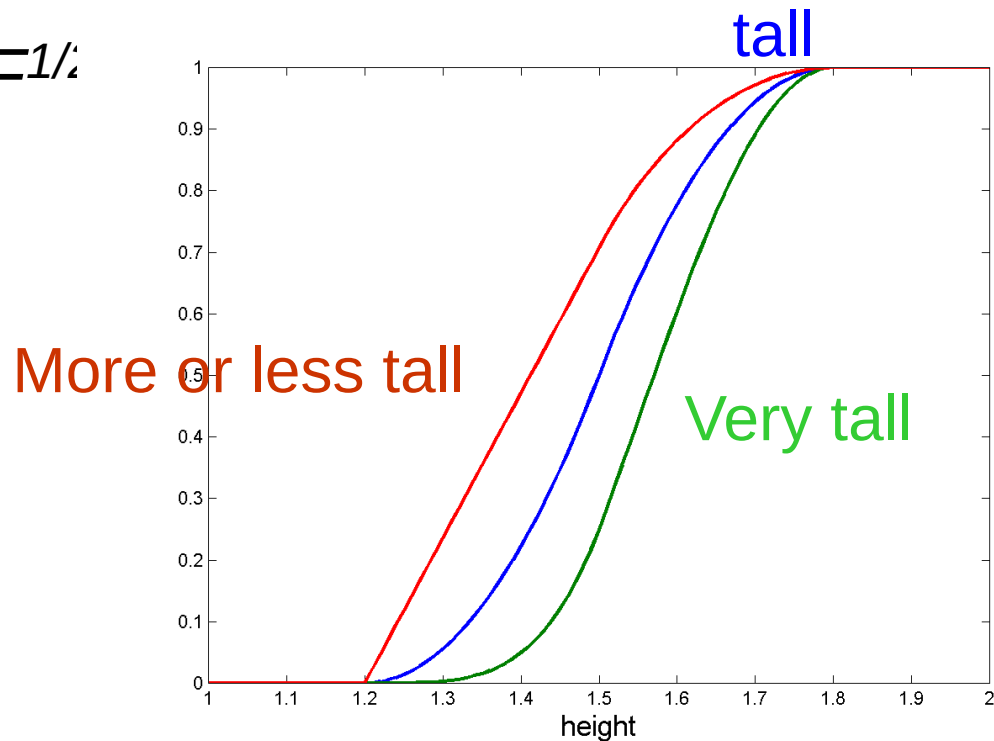
- $\Pi(x, a, b) =$
 - $S(x, b-a, b-a/2, b)$ for $x \leq b$
 - $1 - S(x, b, b+a/2, a+b)$ for $x \geq b$

E.g., **close** (to a)



Linguistic Hedges

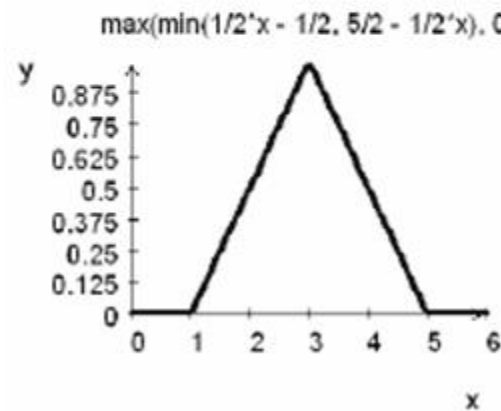
- Modifying the meaning of a fuzzy set using hedges such as *very*, *more or less*, *slightly*, etc.
- *Very* $F = F^2$
- *More or less* $F = F^{1/2}$
- *etc.*



Fuzzy membership function

Triangular MF:

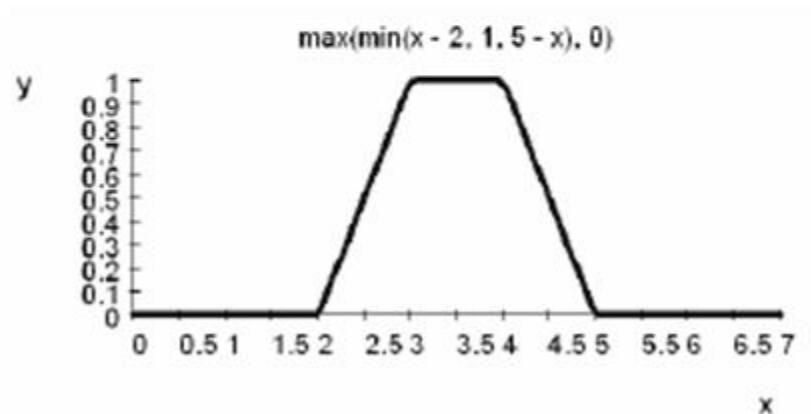
$$\text{triangularmf}(x, c, h) = \max\left(\min\left(\frac{h - c + x}{h}, \frac{c + h - x}{h}\right), 0\right)$$



Fuzzy membership function

Trapezoidal MF:

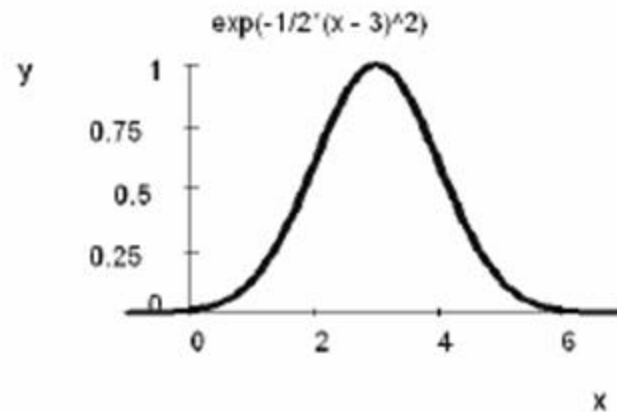
$$\text{trapmf}(x, a, b, c, d) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$



Fuzzy membership function

Gaussian MF:

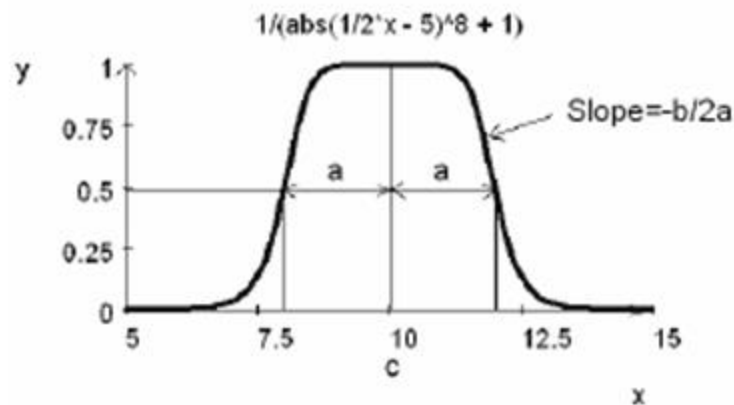
$$\text{gaussmf}(x, c, s) = e^{-\frac{(x-c)^2}{2 \cdot s^2}}$$



Fuzzy membership function

Generalized bell MF:

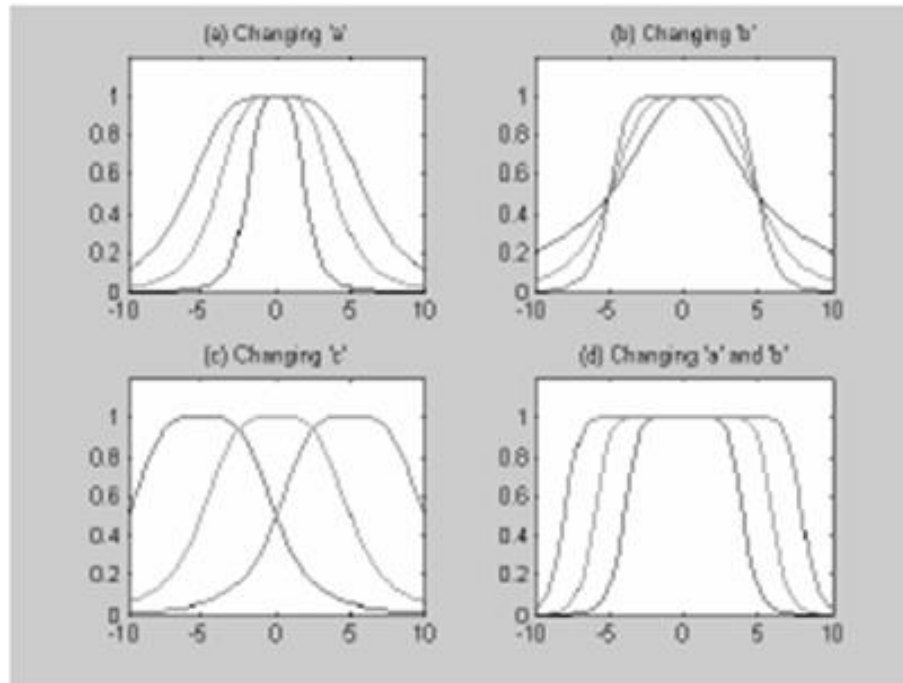
$$gbellmf(x, a, b, c) = \frac{1}{\left| \frac{x}{a} - \frac{c}{a} \right|^{2 \cdot b} + 1}$$



Fuzzy membership function

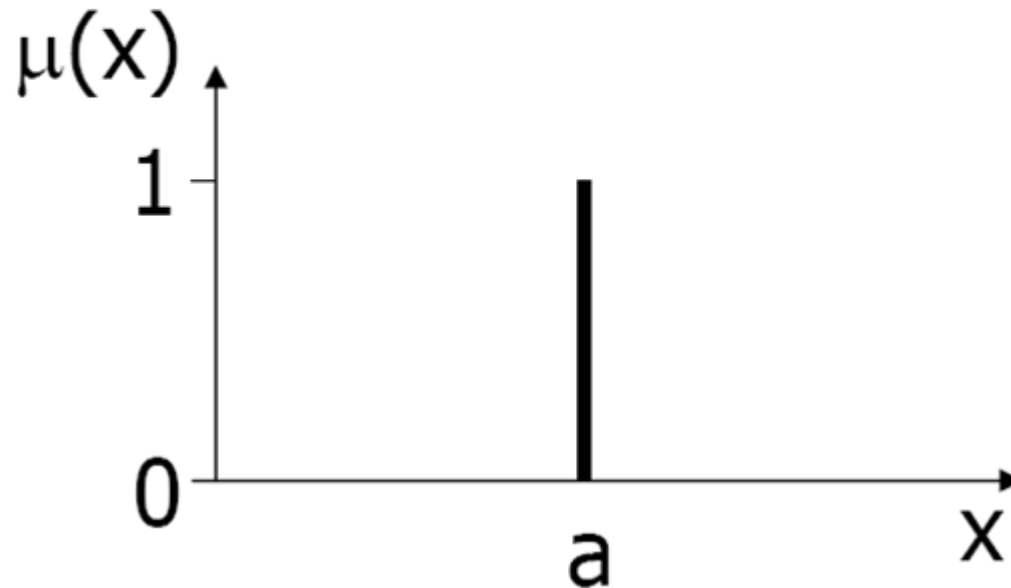
Generalized bell MF:

$$\text{gbellmf}(x, a, b, c) = \frac{1}{\left| \frac{x}{a} - \frac{c}{a} \right|^{2 \cdot b} + 1}$$



Fuzzy membership function

Singleton: (a,1)



Fuzzy membership function

The list of MFs introduced here is by no means exhaustive

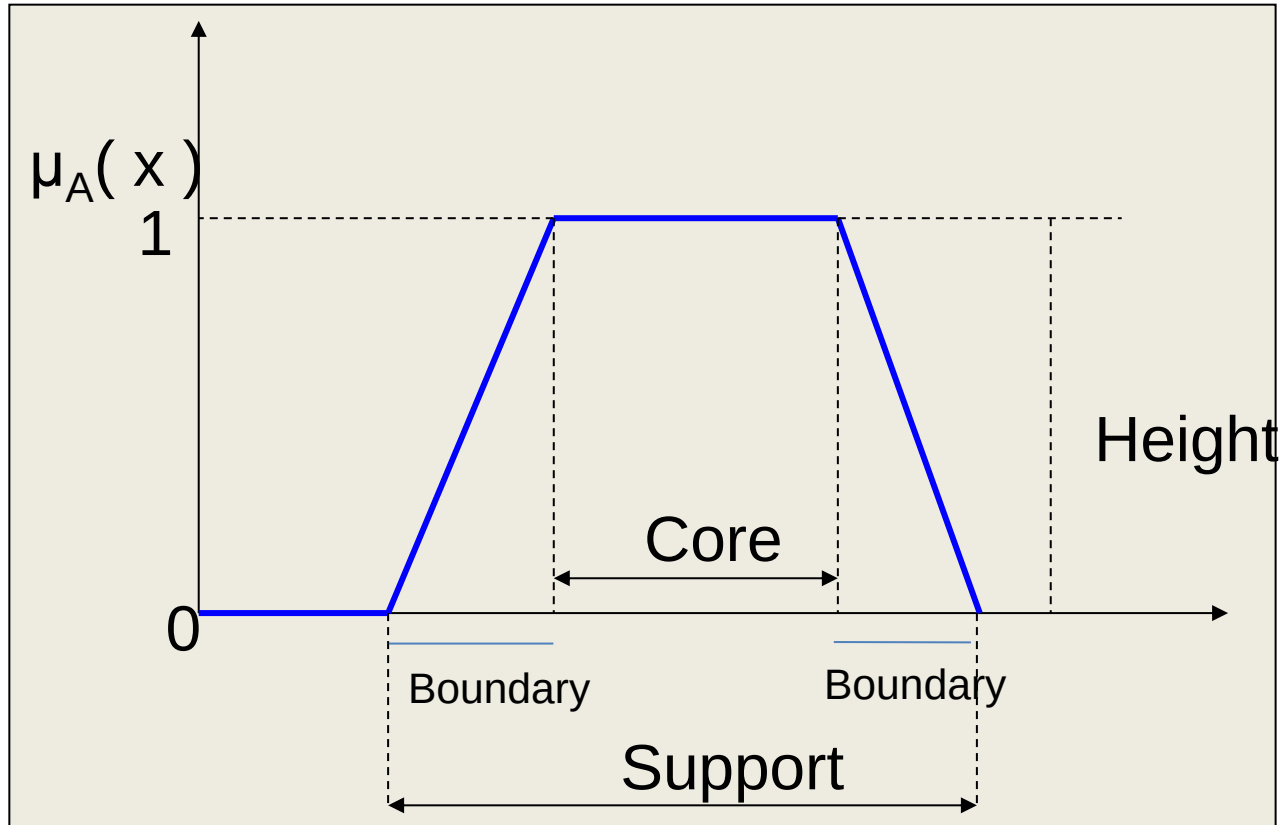
Other specialized MFs can be created for specific applications if necessary

Any type of continuous probability distribution functions can be used as an MF

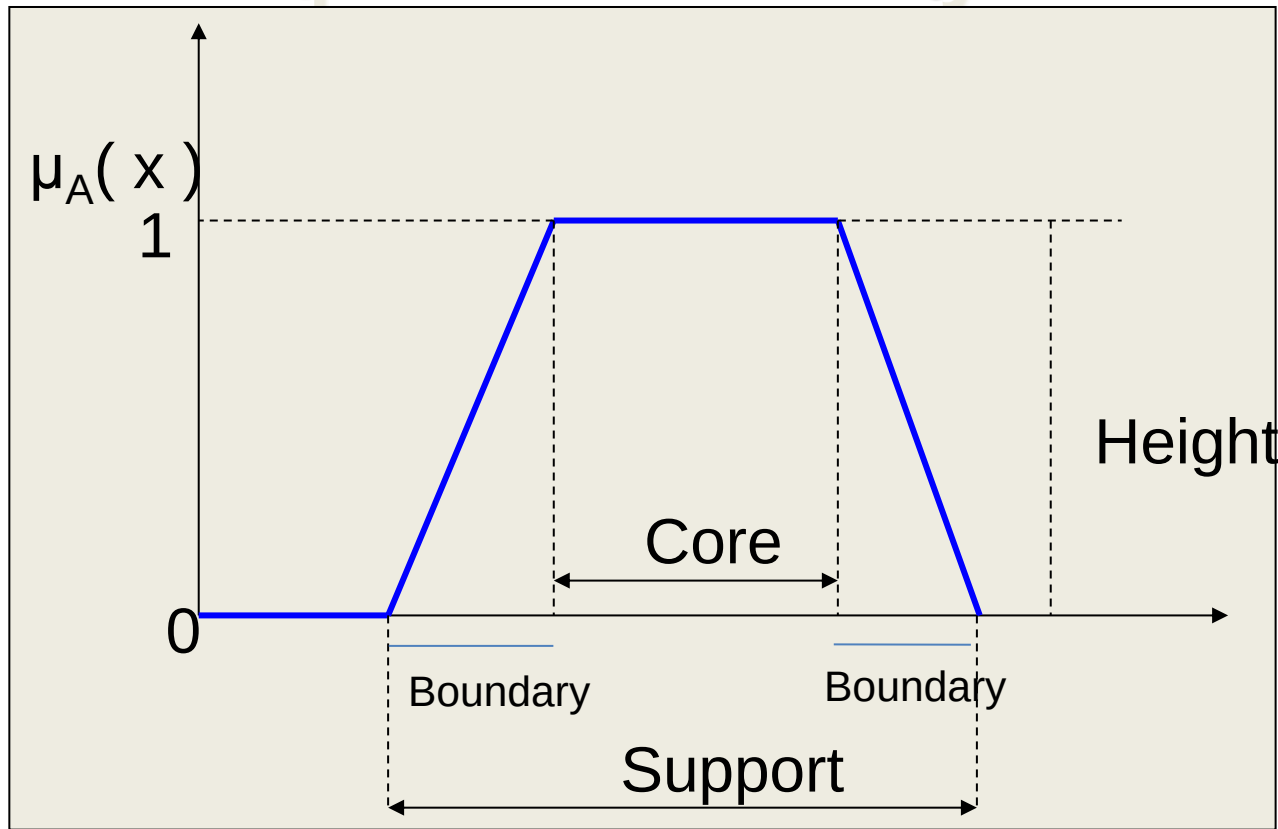
Fuzzy membership function ?

- Subjective evaluation: The shape of the functions is defined by specialists
- Ad-hoc: choose a simple function that is suitable to solve the problem
- Distributions, probabilities: information extracted from measurements
- Adaptation: testing
- Automatic: algorithms used to define functions from data

Properties of Fuzzy set



Properties of Fuzzy set

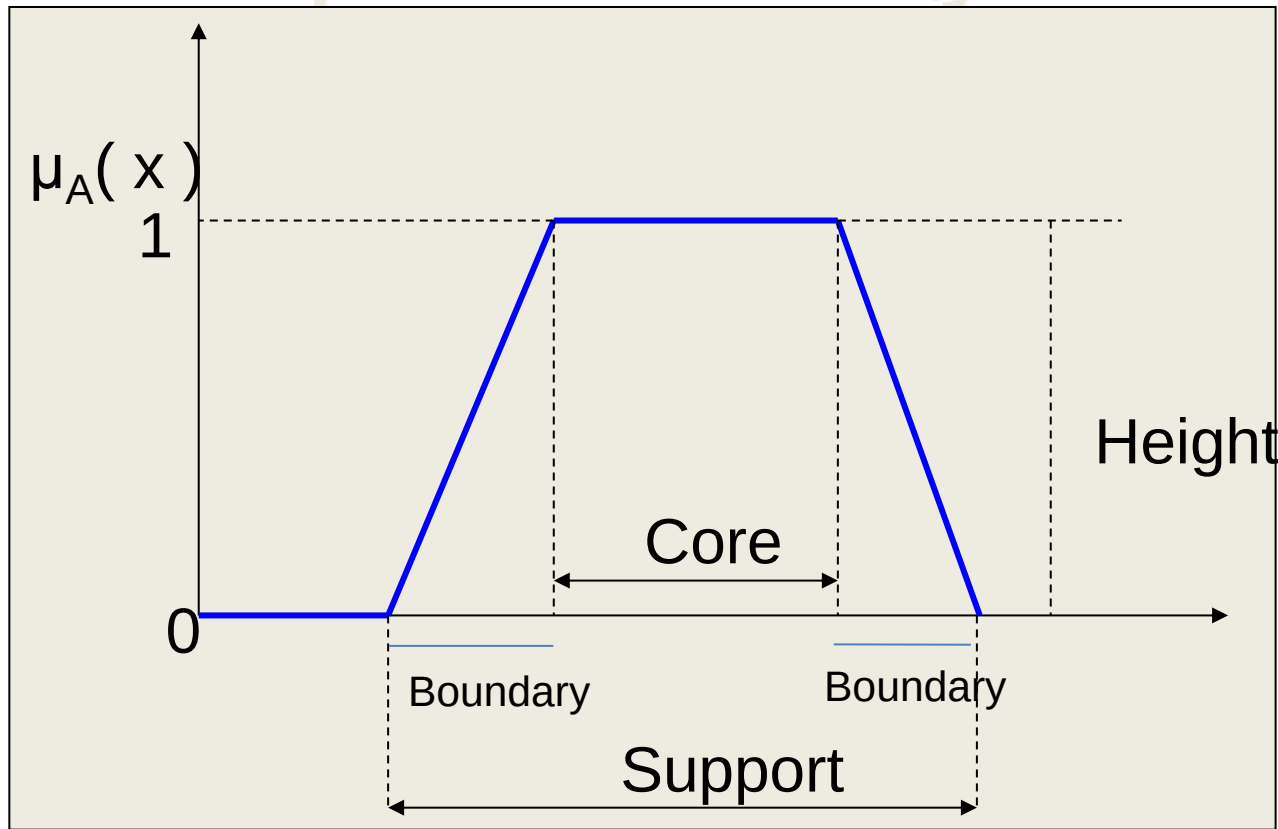


Core: Comprises those elements x of the Universe
Such that:

$$\mu_A(x) = 1$$

$$core(A) = \{x \in X \mid \mu_A(x) = 1\}$$

Properties of Fuzzy set

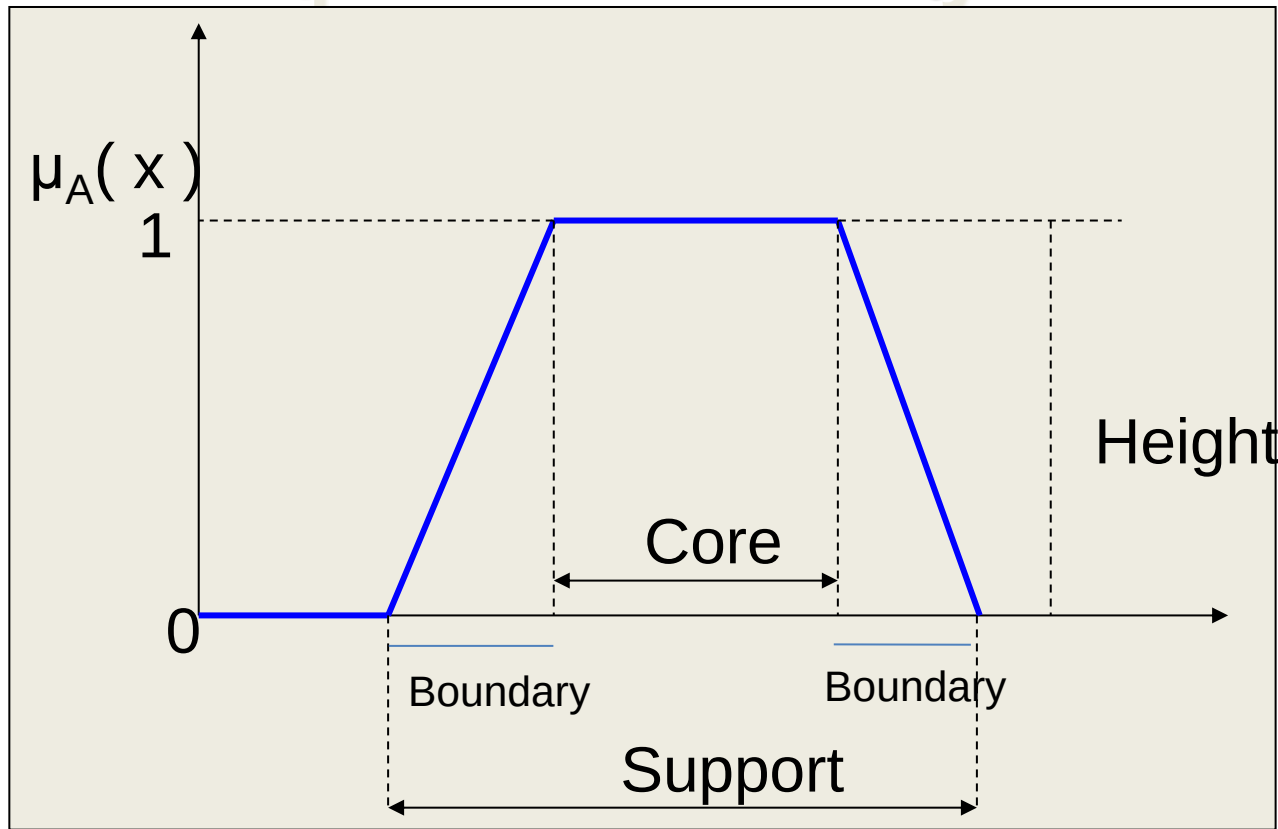


Support: Comprises those elements x of the Universe
Such that:

$$\mu_A(x) > 0$$

$$\text{Support}(A) = \{x \in X \mid \mu_A(x) > 0\}$$

Properties of Fuzzy set



Boundary: Comprises those elements x of the Universe Such that:

$$0 < \mu_A(x) < 1$$

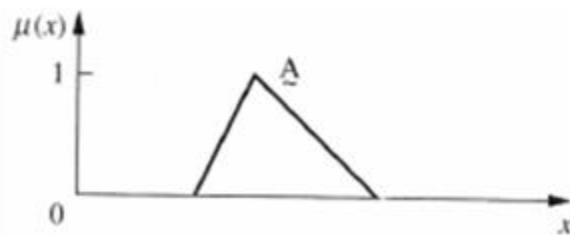
$$\text{Boundary}(A) = \{x \in X \mid 0 < \mu_A(x) < 1\}$$

Properties of Fuzzy set

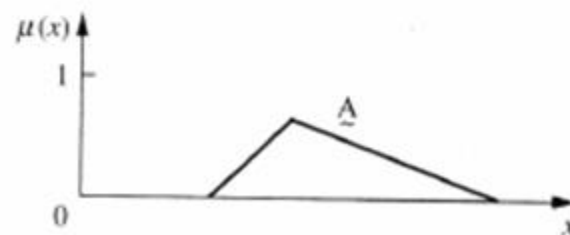
Height : the height of a fuzzy set A is defined as

$$\text{hgt}(A) = \sup_{x \in X} \mu_A(x)$$

Set A is called normal if $\text{hgt}(A)=1$
and subnormal if $\text{hgt}(A)<1$



Normal (a)



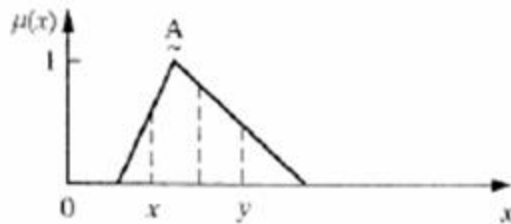
Subnormal (b)

Properties of Fuzzy set

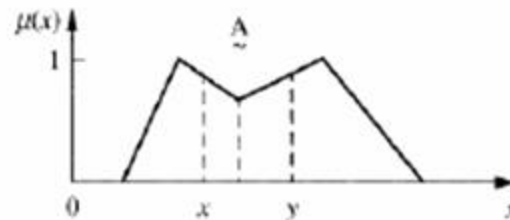
Convex Fuzzy set: a fuzzy set A is convex iff

$$\forall x, y \in X \quad \text{and} \quad \forall \lambda \in [0,1]$$

$$\mu_A(\lambda x + (1-\lambda)y) \geq \min(\mu_A(x), \mu_A(y))$$



Convex, normal fuzzy set

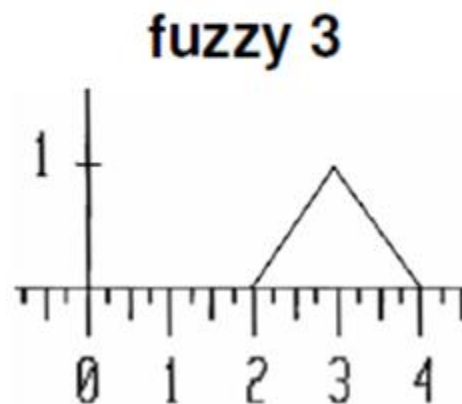


Non convex, normal fuzzy set

Properties of Fuzzy set

Fuzzy number: a fuzzy set A is a fuzzy number if the fuzzy set is

- Convex
- Normal
- The core consists of one value only
- MF is piecewise continuous



Fuzzy Variable

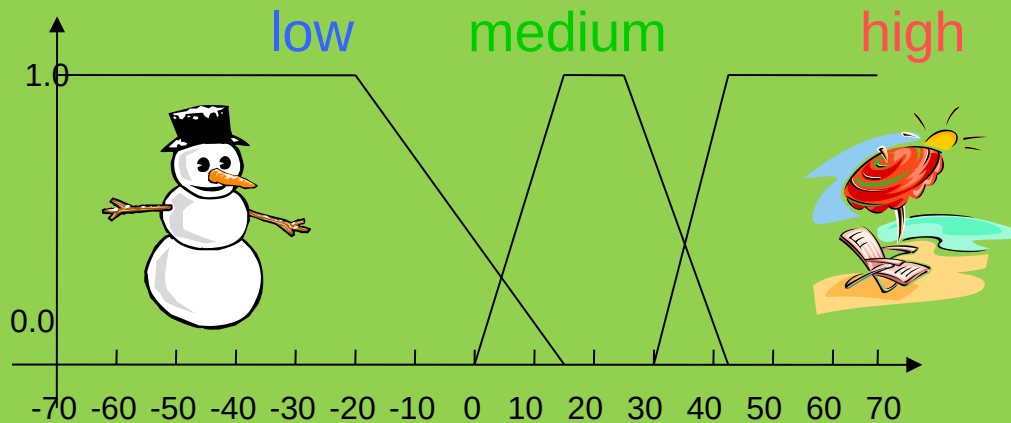
- A fuzzy variable is defined by the quadruple

$$V = \{x, l, u, m\}$$

- X is the variable symbolic name: *temperature*
- L is the set of labels: *low, medium and high*
- U is the **universe of discourse**
- M are the semantic rules that define the meaning of each label in L (membership functions).

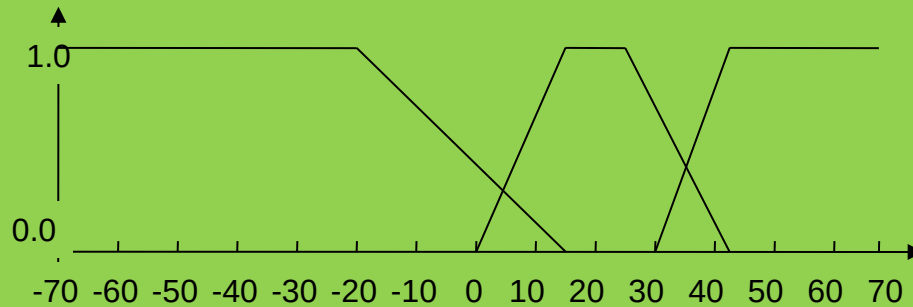
Fuzzy Variable Example

- X = Temperature
- $L = \{\text{low, medium, high}\}$
- $U = \{x \in X \mid -70^\circ \leq x \leq +70^\circ\}$
- $M =$

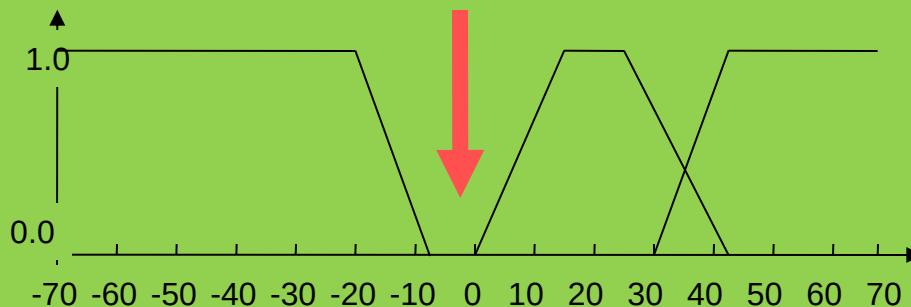


Fuzzy Variable

- Completeness: A variable is complete if for any $x \in X$ there is a fuzzy set such as $\mu(x) > 0$

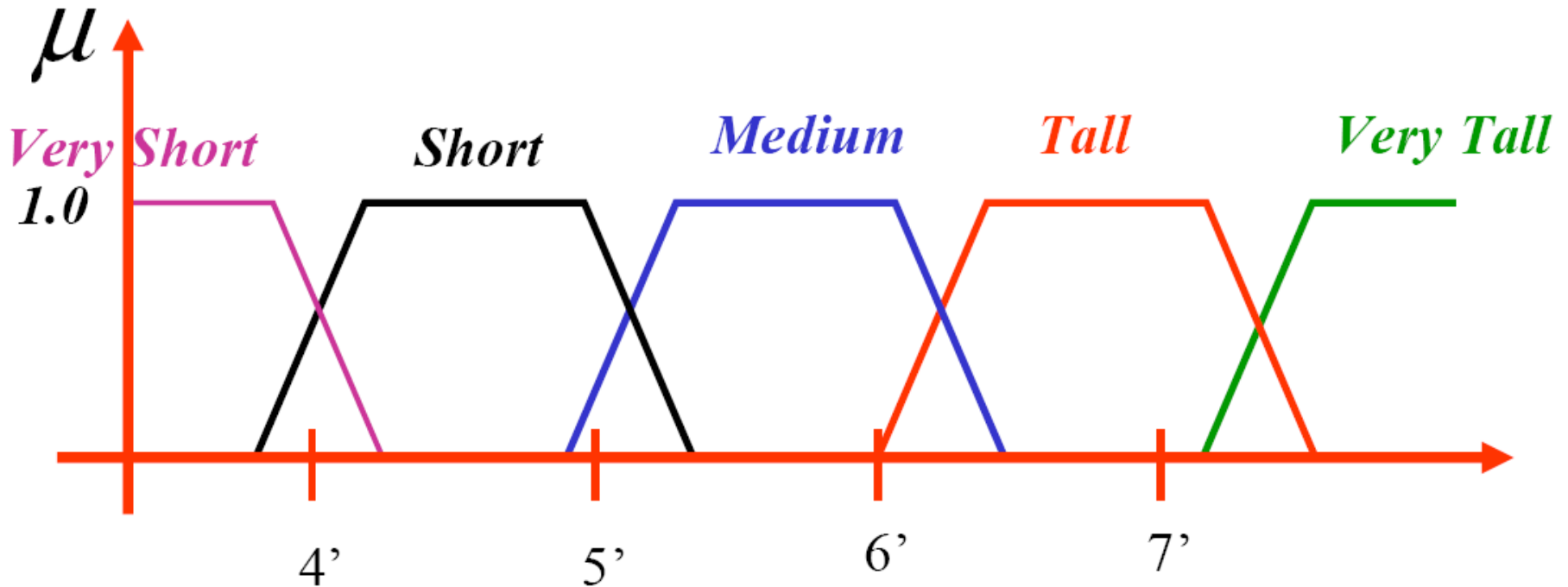


Complete



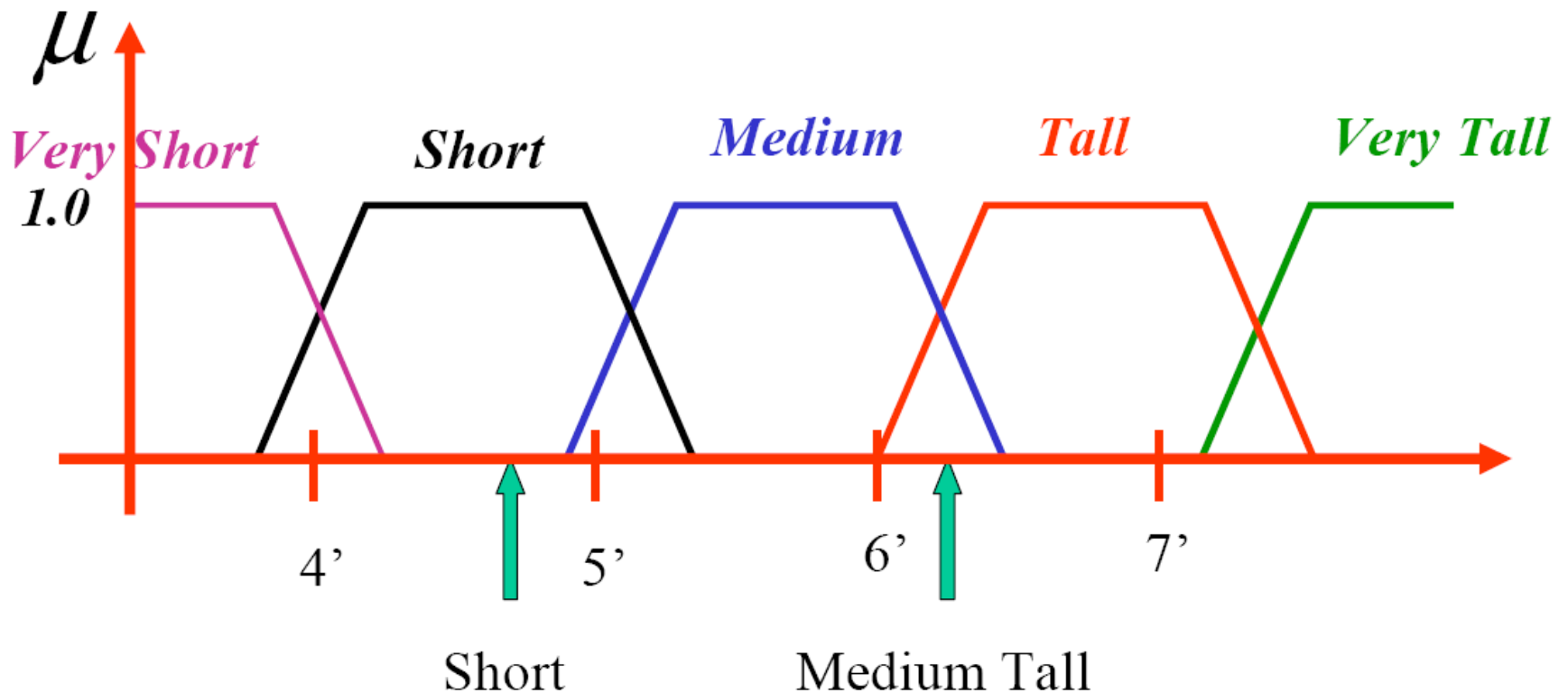
Incomplete

Fuzzy membership function



$$\mu = [\mu_{vs}, \mu_s, \mu_m, \mu_t, \mu_{vt}]$$

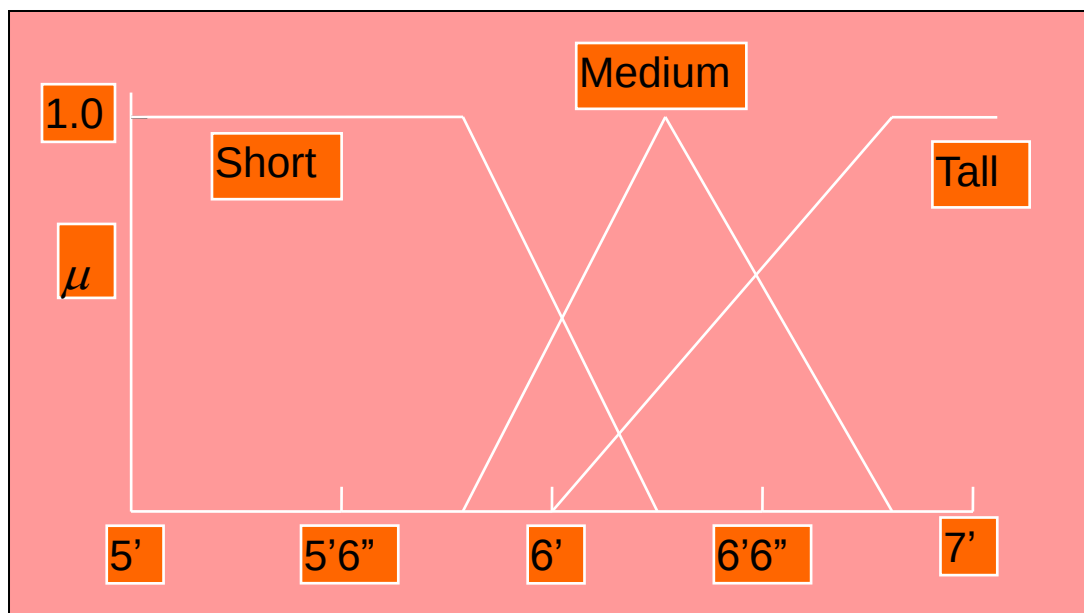
Fuzzy membership function



$$\mu = [0, 1, 0, 0, 0]$$

$$\mu = [0, 0, 0.5, 0.5, 0]$$

Example: Short, Medium height and Tall



Note: $\mu_{\text{Short}}(x) + \mu_{\text{Medium}}(x) + \mu_{\text{Tall}}(x) \neq 1$.

Someone who is 6'1" is simultaneously short, of medium height

and tall in various degrees which don't add to 1.

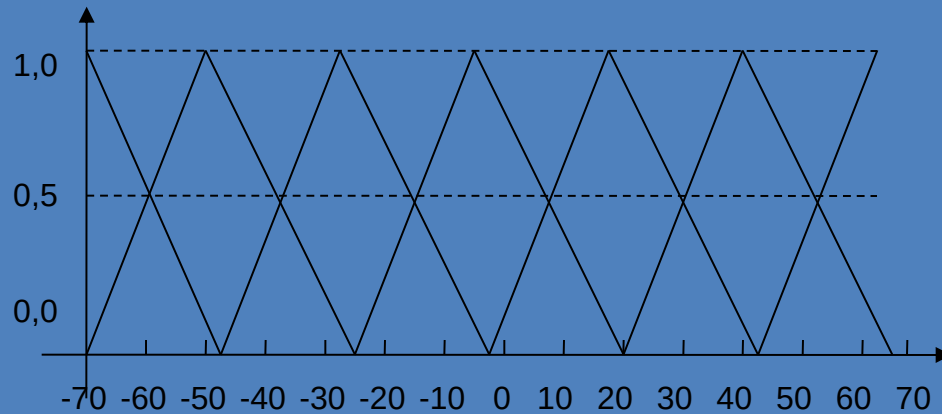
Partition of Unity

- A fuzzy variable forms a partition of unity if for each input value x

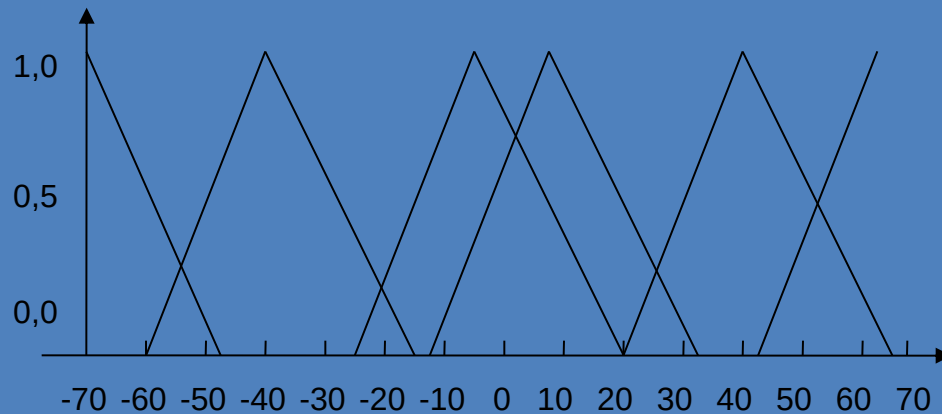
$$\sum_{i=1}^p \mu_{A_i}(x) \equiv 1$$

- where p is the number of sets to which x belongs
- There is no rule to define the overlapping degree between two neighbouring sets
- A rule of thumb is to use 25% to 50%

Partition of Unity



Partition of
Unity



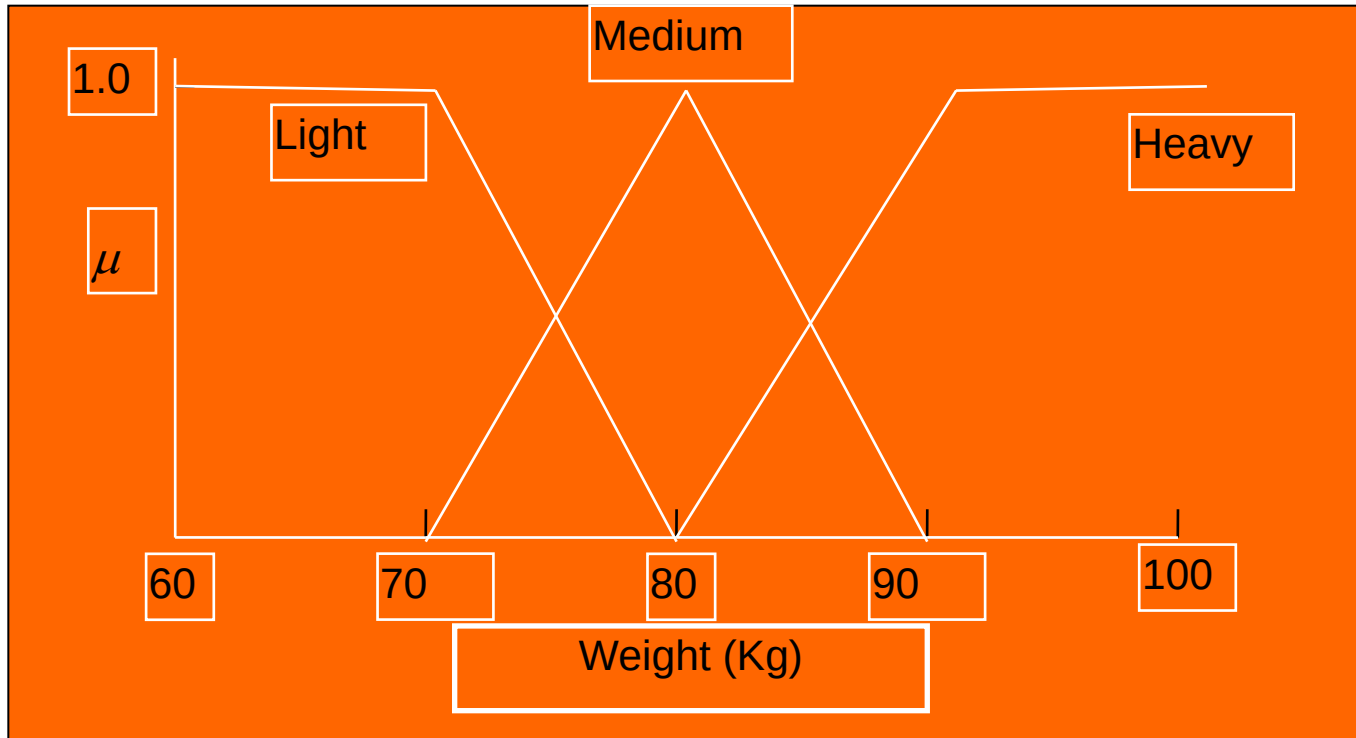
No Partition
of
Unity

Partition of Unity cont

- Any complete fuzzy variable may be transformed into a partition of unity using the equation

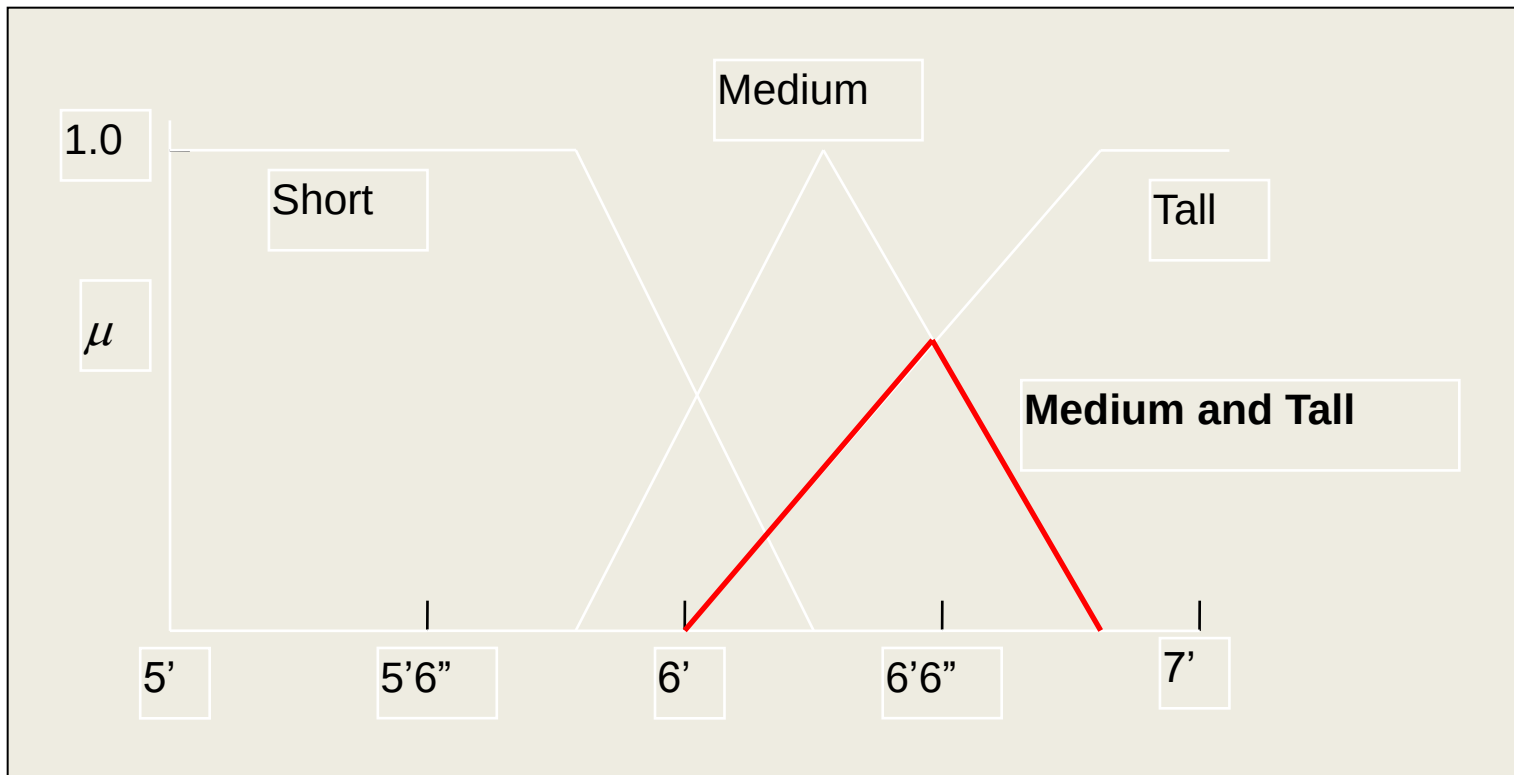
$$\mu_{\hat{A}_i}(x) = \frac{\mu_{A_i}(x)}{\sum_{j=1}^p \mu_{A_j}(x)} \text{ for } i = 1, \dots, p$$

Example: Light, Medium weight and Heavy



$$\mu_{A.B}(x) = \min(\mu_A(x), \mu_B(x))$$

- Example: *Medium and Tall*



Note: No one is a full member of this set.

Fuzzy Logic Operations

Fuzzy union operation or fuzzy *OR* :

$$\mu_{A+B}(x) = \max(\mu_A(x), \mu_B(x))$$

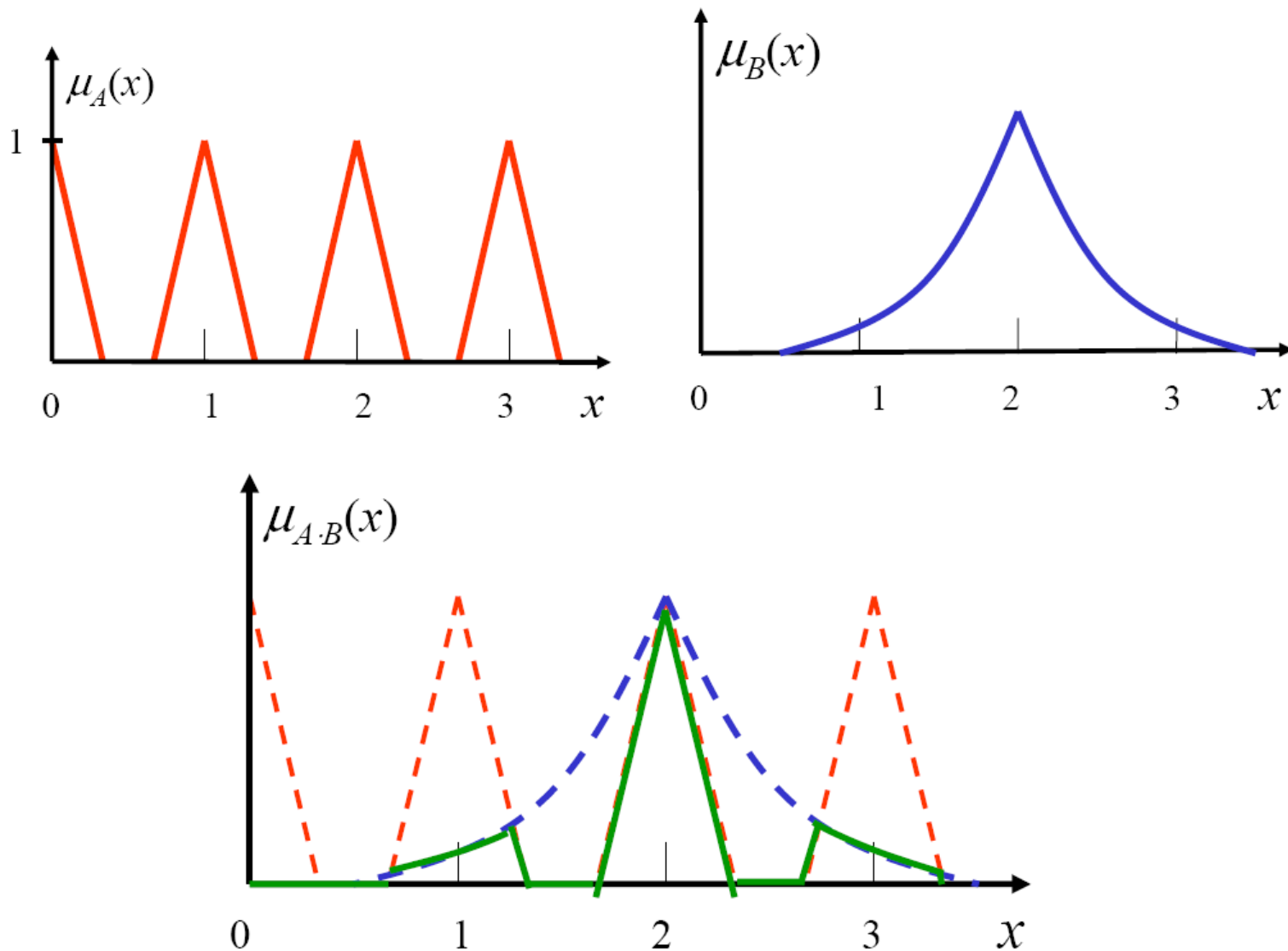
Fuzzy intersection operation or fuzzy *AND*

$$\mu_{A.B}(x) = \min(\mu_A(x), \mu_B(x))$$

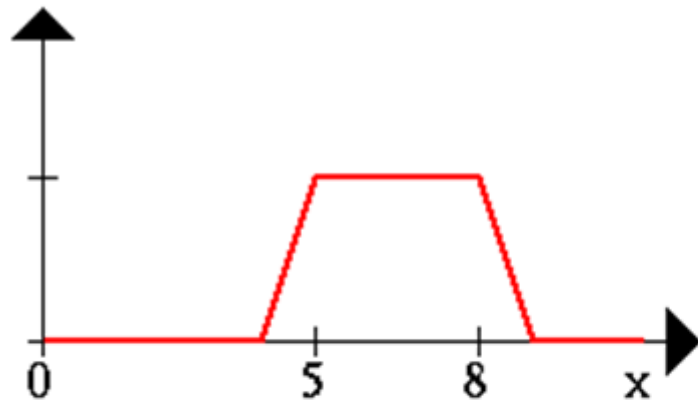
Complement operation

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

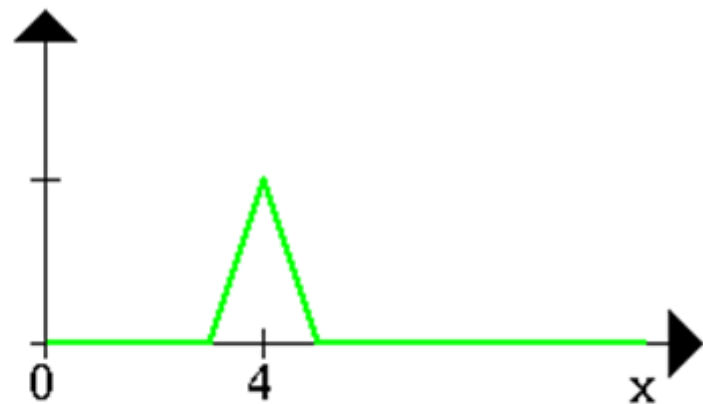
$$\mu_{A.B}(x) = \min(\mu_A(x), \mu_B(x))$$



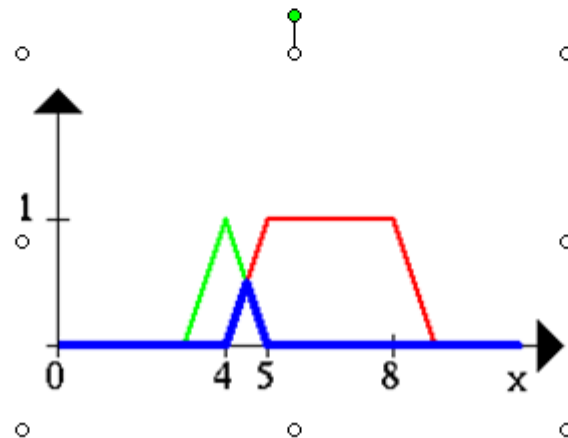
$$\mu_{A.B}(x) = \min(\mu_A(x), \mu_B(x))$$



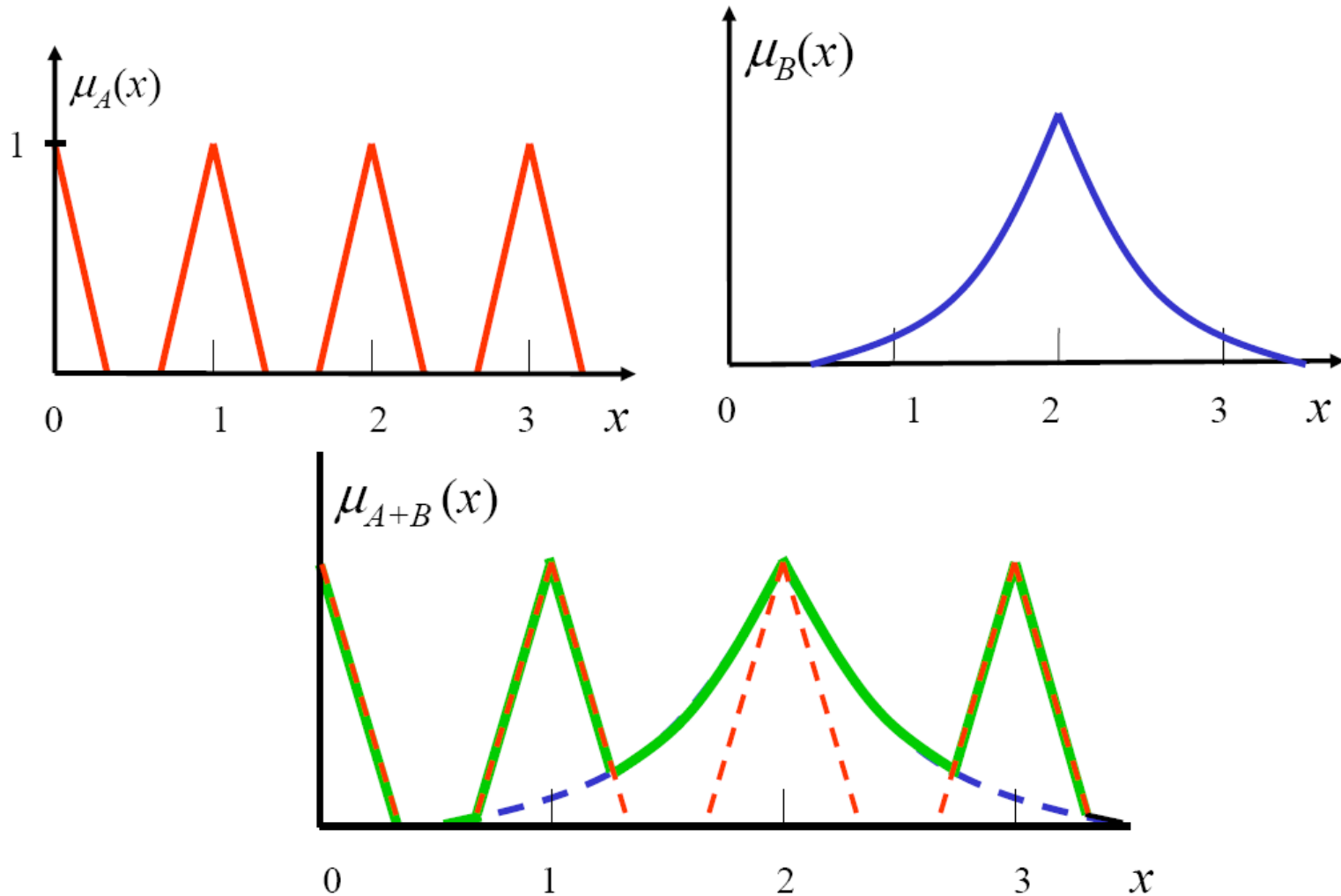
A



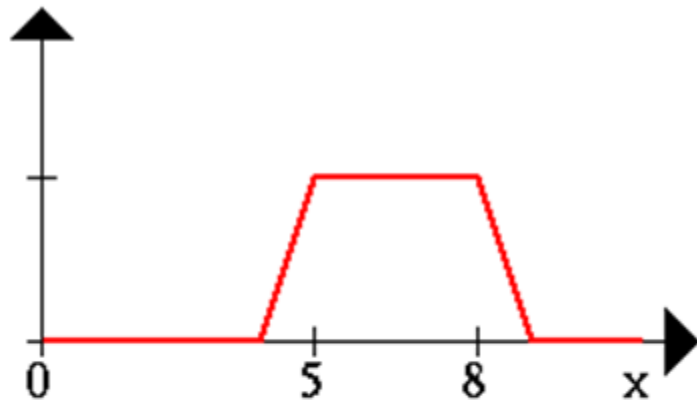
B



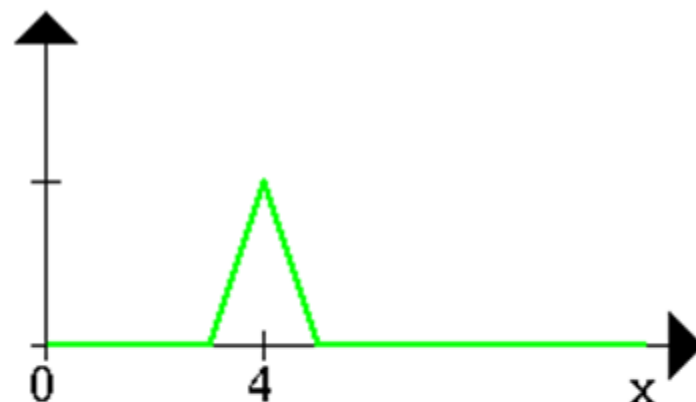
$$\mu_{A+B}(x) = \max(\mu_A(x), \mu_B(x))$$



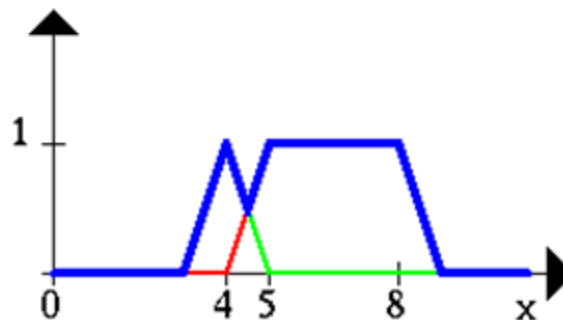
$$\mu_{A+B}(x) = \max(\mu_A(x), \mu_B(x))$$



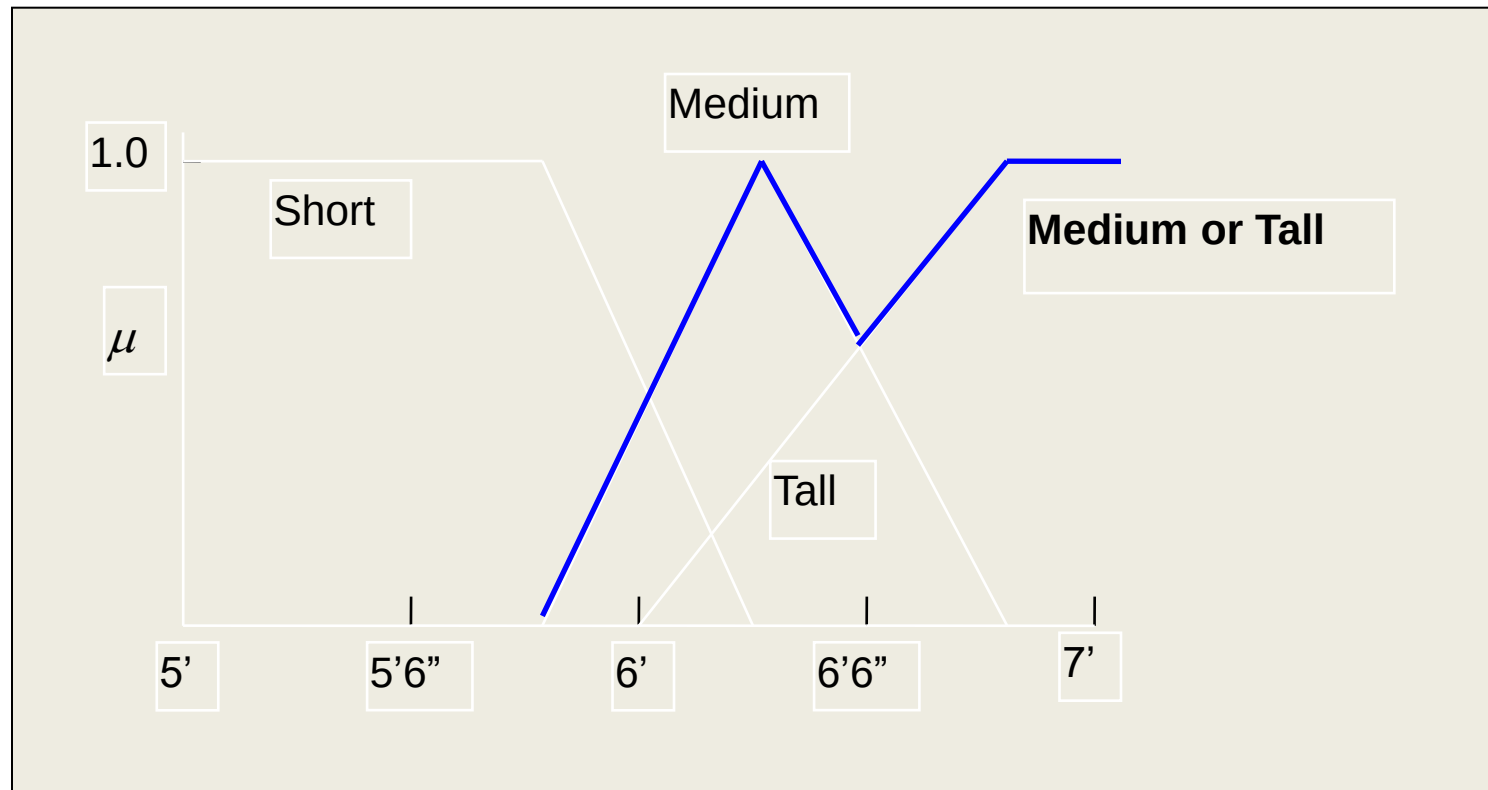
A



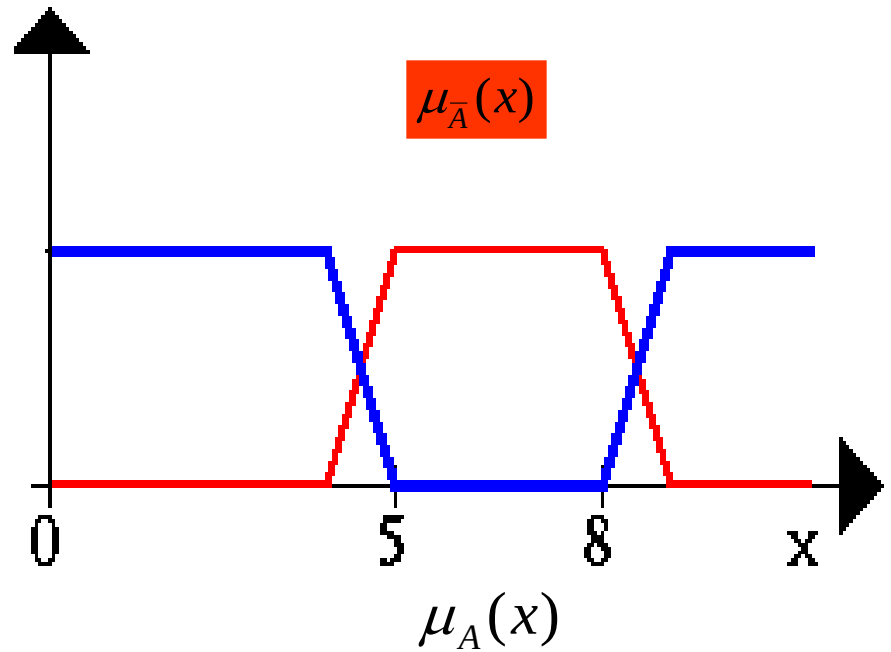
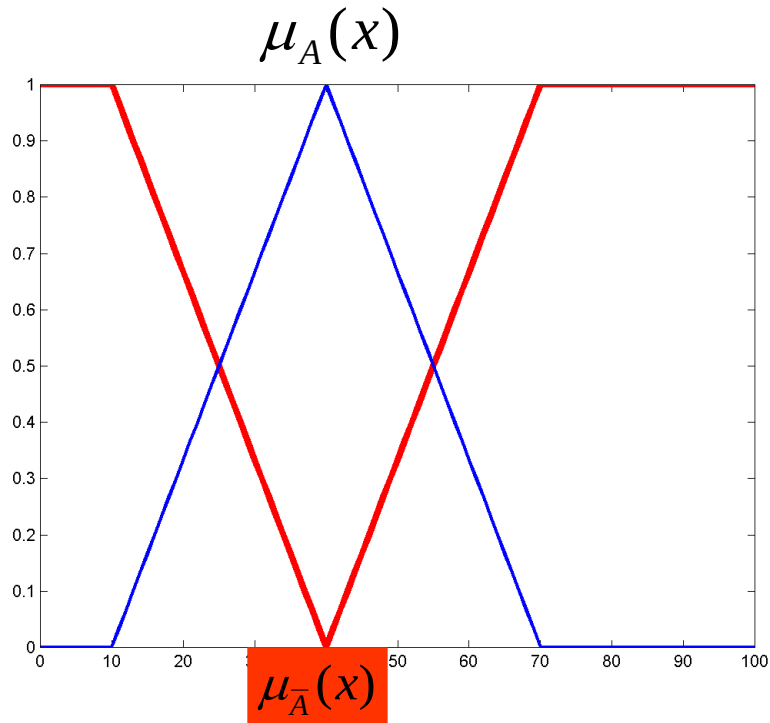
B



$$\mu_{A+B}(x) = \max(\mu_A(x), \mu_B(x))$$



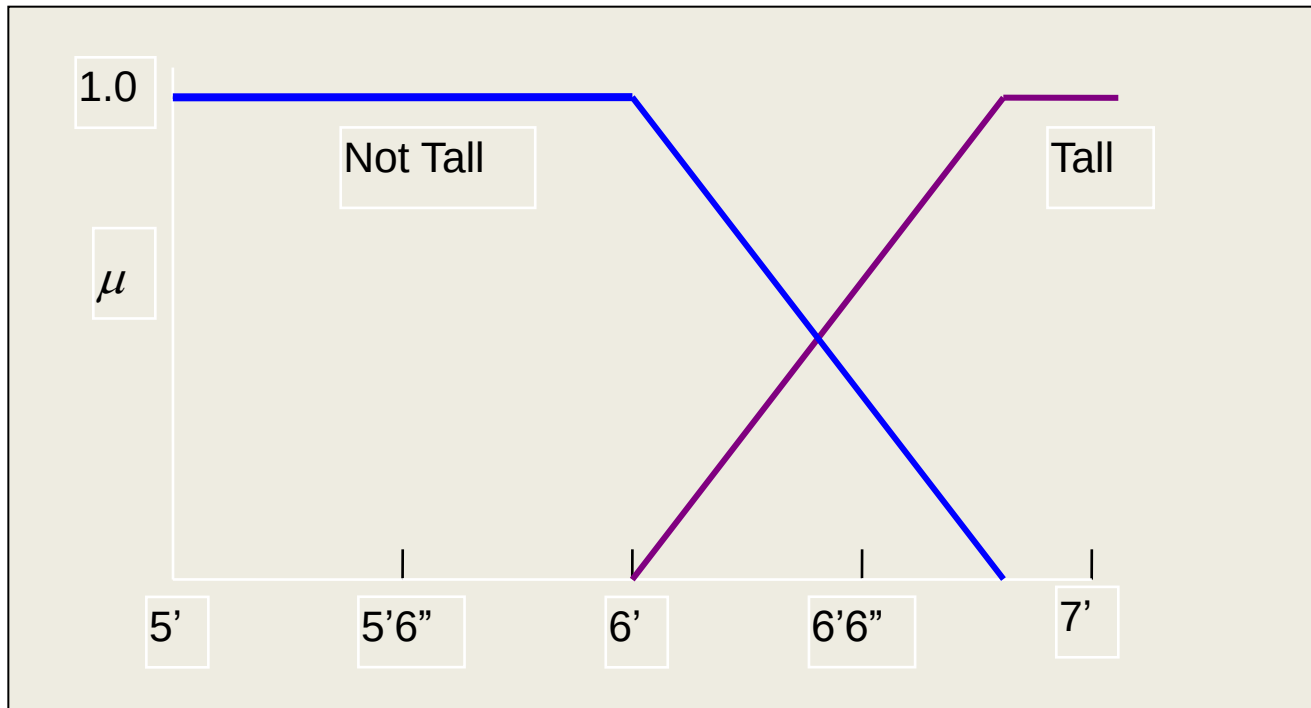
$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$



$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

Negation:

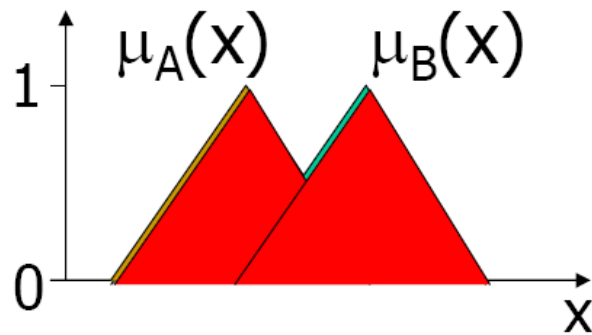
$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$



Fuzzy Logic Operations

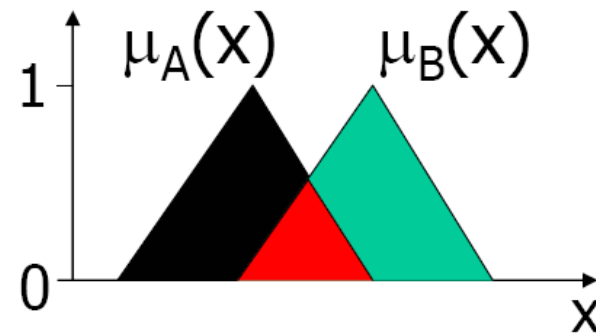
Union

$$\mu_{A \vee B}(x) = \max\{\mu_A(x), \mu_B(x)\}$$

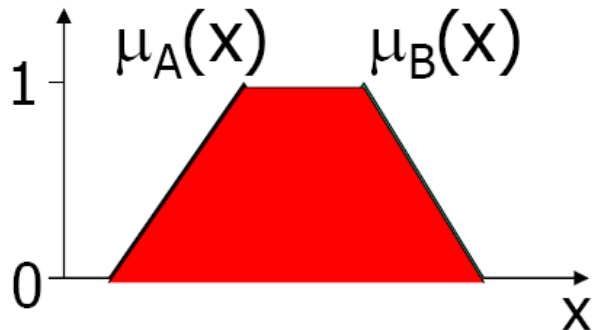


Intersection

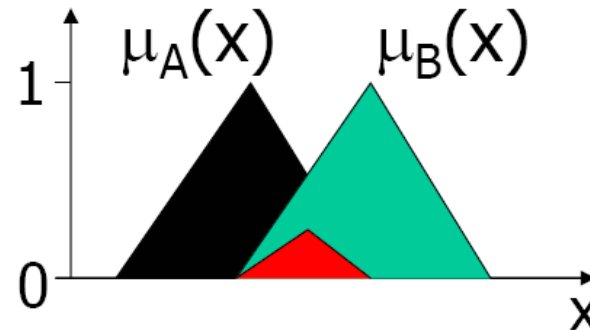
$$\mu_{A \wedge B}(x) = \min\{\mu_A(x), \mu_B(x)\}$$



$$\mu_{A \vee B}(x) = \min\{1, \mu_A(x) + \mu_B(x)\}$$



$$\mu_{A \wedge B}(x) = \mu_A(x) \cdot \mu_B(x)$$

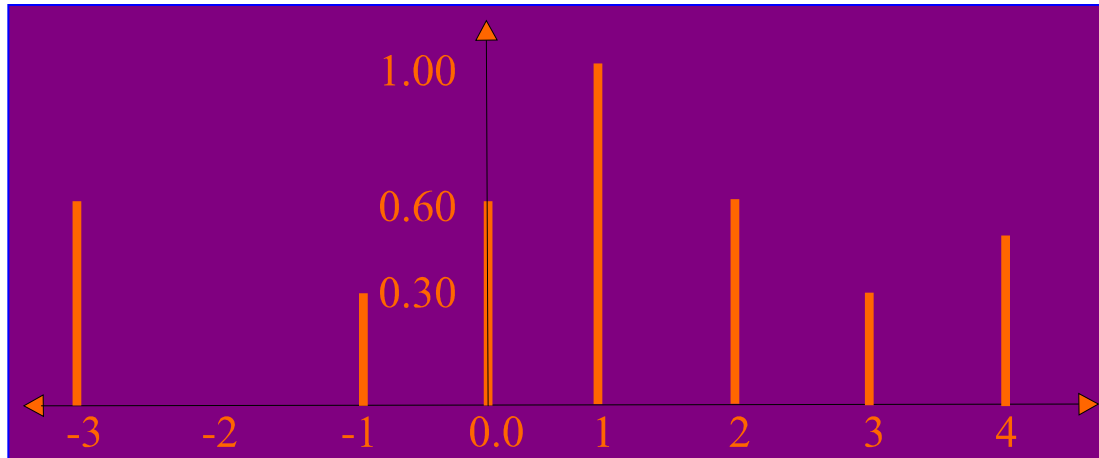


Discrete fuzzy subset

A simple example of a discrete universe and a discrete fuzzy subset of it, is:

$$X = \{-3, -2, -1, 0, 1, 2, 3, 4\}$$

$$A = 0.6/-3 + 0.0/-2 + 0.3/-1 + 0.6/0 + 1.0/1 + 0.6/2 + 0.3/3 + 0.5/4$$



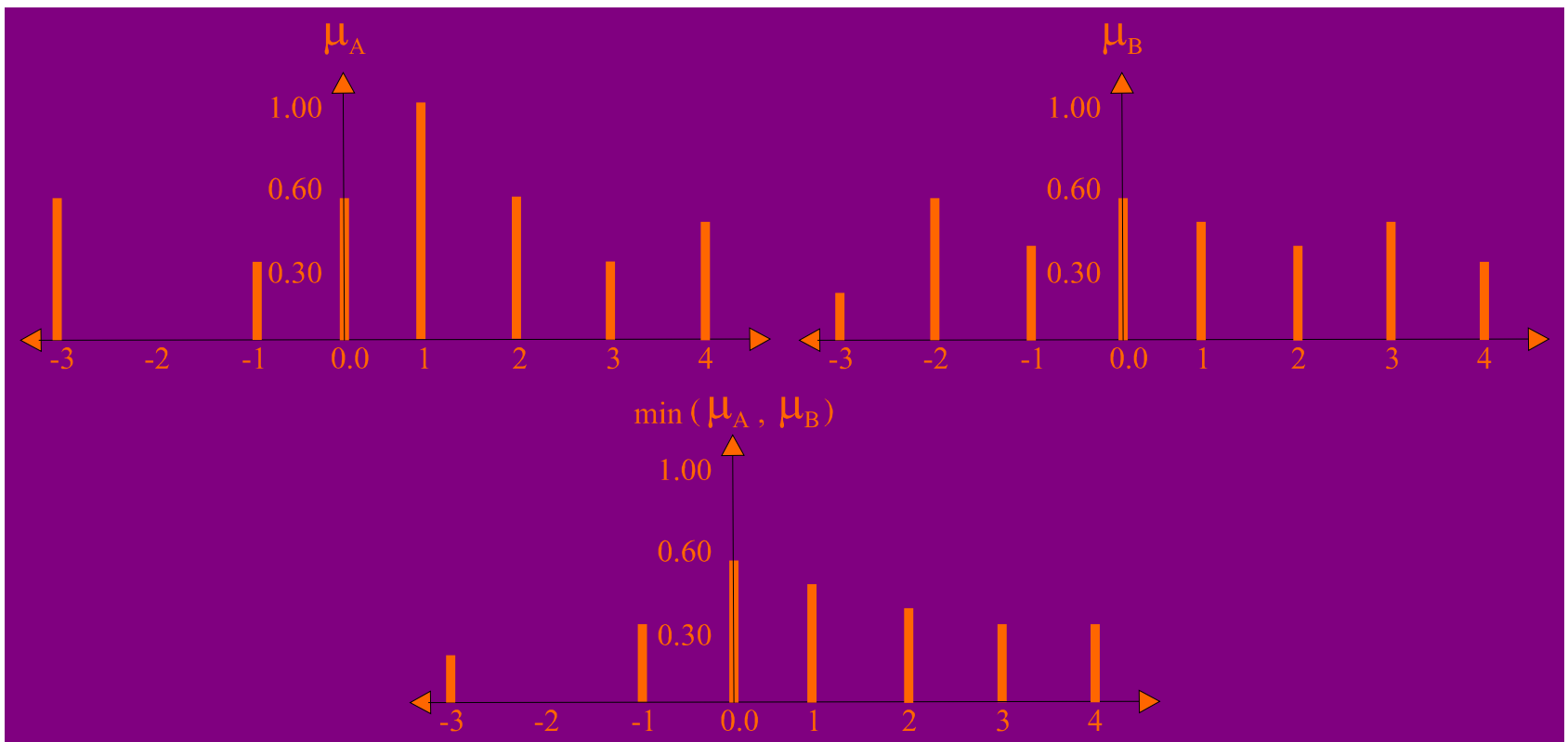
$$\mu_{A.B}(x) = \min(\mu_A(x), \mu_B(x))$$

Let A and B be fuzzy subsets of the universe $X = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

$A = 0.6/-3 + 0.0/-2 + 0.3/-1 + 0.6/0.0 + 1.0/1 + 0.6/2 + 0.3/3 + 0.5/4$

$B = 0.2/-3 + 0.6/-2 + 0.4/-1 + 0.6/0.0 + 0.5/1 + 0.4/2 + 0.5/3 + 0.3/4$

$A \cap B = 0.2/-3 + 0.0/-2 + 0.3/-1 + 0.6/0.0 + 0.5/1 + 0.4/2 + 0.3/3 + 0.3/4$



Possible operators for AND in fuzzy logic

$$\mu_A(X) \cdot \mu_B(X)$$

$$\frac{\mu_A(X) \cdot \mu_B(X)}{\mu_A(X) + \mu_B(X) - \mu_A(X) \cdot \mu_B(X)}$$

$$\max\{0, \mu_A(X) + \mu_B(X) - 1\}$$

$$\frac{\mu_A(X) \cdot \mu_B(X)}{2 - [\mu_A(X) + \mu_B(X) - \mu_A(X) \cdot \mu_B(X)]}$$

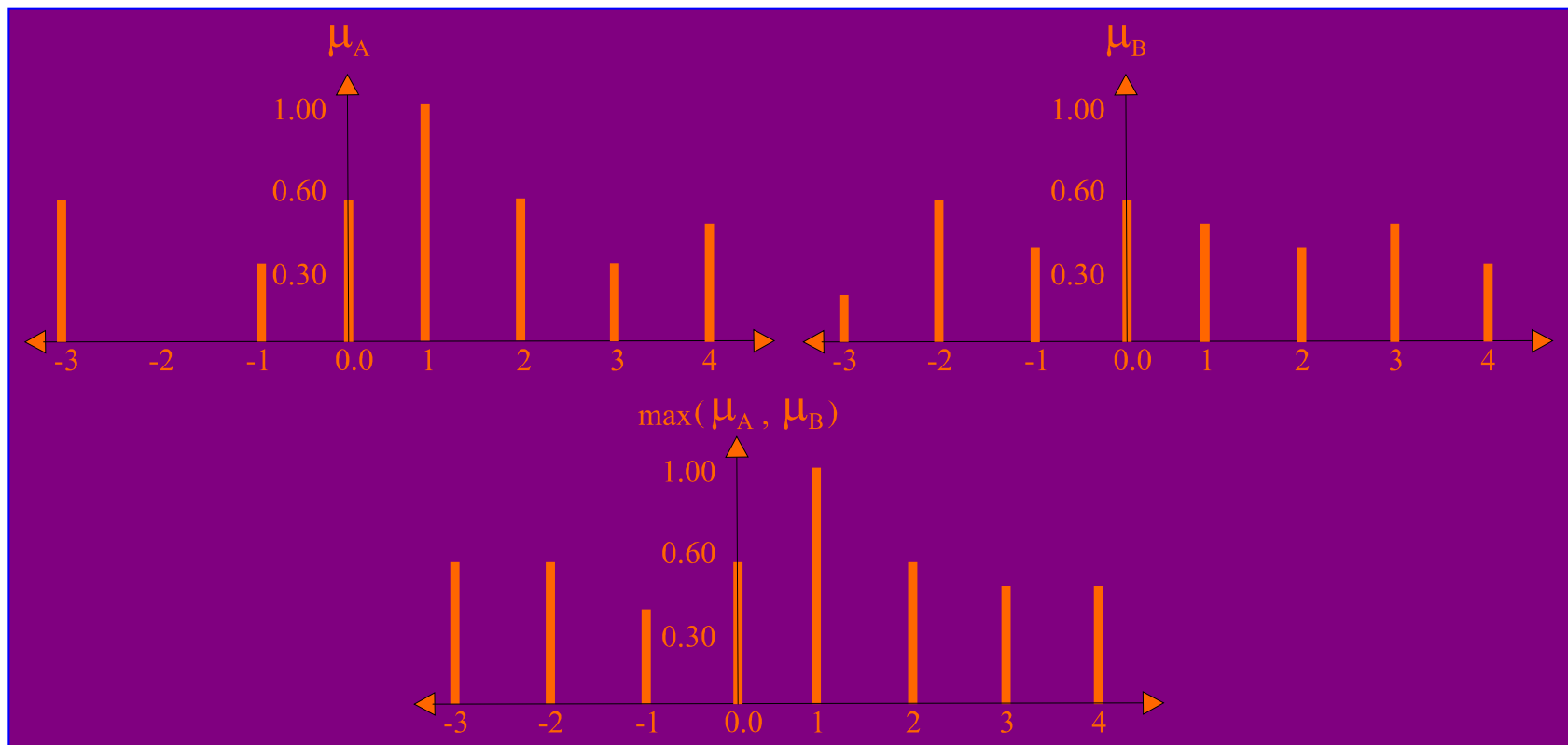
$$\mu_{A+B}(x) = \max(\mu_A(x), \mu_B(x))$$

Let A and B be fuzzy subsets of the universe $X = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

$A = 0.6/-3 + 0.0/-2 + 0.3/-1 + 0.6/0.0 + 1.0/1 + 0.6/2 + 0.3/3 + 0.5/4$

$B = 0.2/-3 + 0.6/-2 + 0.4/-1 + 0.6/0.0 + 0.5/1 + 0.4/2 + 0.5/3 + 0.3/4$

$A \cup B = 0.6/-3 + 0.6/-2 + 0.4/-1 + 0.6/0.0 + 1.0/1 + 0.6/2 + 0.5/3 + 0.5/4$



Possible operators for OR in fuzzy logic

$$\mu_A(X) + \mu_B(X) - \mu_A(X) \cdot \mu_B(X)$$

$$\frac{\mu_A(X) + \mu_B(X) - 2\mu_A(X) \cdot \mu_B(X)}{1 - \mu_A(X) \cdot \mu_B(X)}$$

$$\frac{\mu_A(X) + \mu_B(X)}{1 + \mu_A(X) \cdot \mu_B(X)}$$

$$\min\{1, \mu_A(X) + \mu_B(X)\}$$

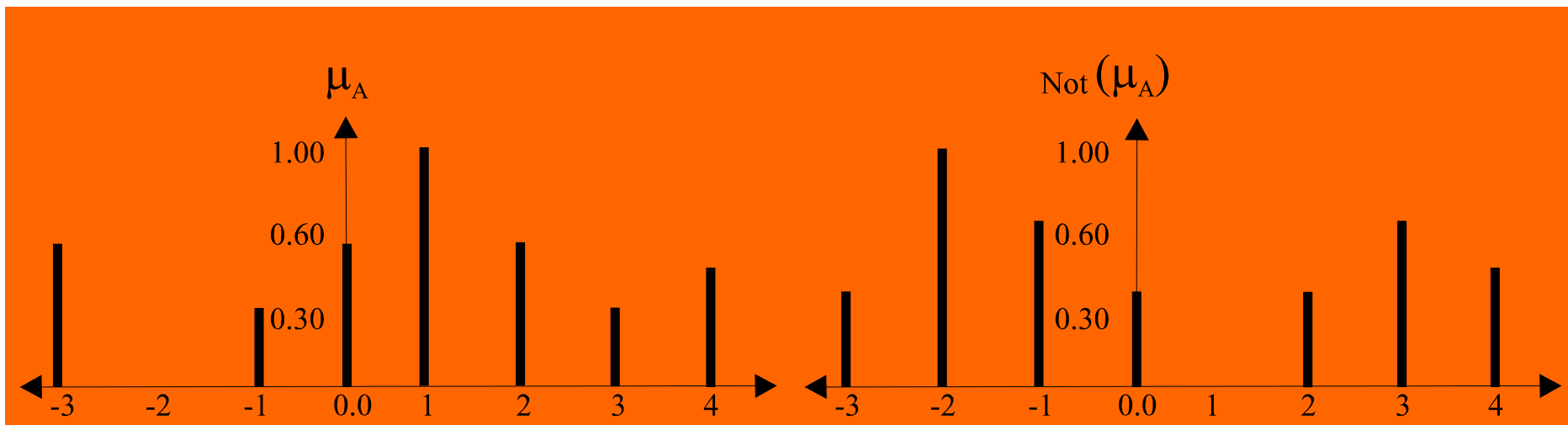
$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

Let A be fuzzy subset of the universe

$X = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

$$A = 0.6/-3 + 0.0/-2 + 0.3/-1 + 0.6/0.0 + 1.0/1 + 0.6/2 + 0.3/3 + 0.5/4$$

$$\bar{A} = 0.4/-3 + 1.0/-2 + 0.7/-1 + 0.4/0.0 + 0.0/1 + 0.4/2 + 0.7/3 + 0.5/4$$



Questions? **Discussion?** Suggestions ?





Thank you
Thank you